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On the common transversal probability

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Abstract. Let G be a finite group, and let H be a subgroup of G. We compute the probability, denoted by $P_G(H)$, that a left transversal of H in G is also a right transversal, thus a two-sided one. Moreover, we define, and denote by $\mathrm{tp}(G)$, the common transversal probability of G to be the minimum, taken over all subgroups H of G, of $P_G(H)$. We prove a number of results regarding the invariant $\mathrm{tp}(G)$, like lower and upper bounds, and possible values it can attain. We also show that $\mathrm{tp}(G)$ determines structural properties of G. Finally, several open problems are formulated and discussed.

1 Introduction

We begin with some general remarks and a little bit of history. It is often the case that probability theory interacts with group theory. The area that we now call "probabilistic group theory" may reasonably be said to begin with a series of papers by Erdős and Turán – stretching from 1965 to 1972 and beginning with [7] – where statistical properties of the symmetric group are examined in detail. For survey articles of this area, we refer the interested reader to [3, 5], as well as the recent [18]. Prominent concepts that probabilistic group theory concerns itself with include the so-called "probabilistic generation" and the "commuting probability" as well as variants of the latter. The former is about questions of the following type:

> given a family of groups $\{G_n\}_n$, what is the probability that a d-tuple of elements from G_n, chosen uniformly at random, generates G_n as $n \to \infty$?

Recent work in this direction has mainly focussed on the so-called $(2, 3)$-generation problem for finite simple groups. The question here is whether all finite simple groups can be generated by an involution and an element of order 3.

The commuting probability of G, often denoted by $\mathrm{cp}(G)$, is the probability that two elements of G, chosen uniformly at random, commute. It was popularised by Gustafson [12] who, in turn, traces its origin to Erdős and Turán's series of papers

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on the statistics of the symmetric group. Using the class equation, one can prove that $\mathrm{cp}(G) = k(G)/|G|$, where $k(G)$ is the number of conjugacy classes of G. Gustafson also established the gap result stating that, if G is non-abelian, then $\mathrm{cp}(G) \leqslant 5/8$, with equality if and only if $G/Z(G) \cong C_2 \times C_2$. Of course, G is abelian if and only if $\mathrm{cp}(G) = 1$ and, despite the $(5/8, 1)$ gap, the commuting probability may be viewed as an arithmetic quantification of the "abelianness" of a group.

Most research on the commuting probability has focussed on obtaining inequalities for $\mathrm{cp}(G)$ in terms of other invariants of the group and proving that if $\mathrm{cp}(G) > \varepsilon$, then G is in some sense close to abelian. The interested reader is referred to the paper [10] of Guralnick and Robinson, which contains a wealth of information on the commuting probability, and is rich in novel results. One of their most striking results is that $\mathrm{cp}(G) \to 0$ as either the index or the derived length of the Fitting subgroup of G tends to infinity. One can therefore deduce at once that $\mathrm{cp}(G) \to 0$ as $|G| \to \infty$, where G is a non-abelian finite simple group. Finally, we direct the reader's attention to the recent paper by Eberhard [6], which answers in the affirmative some of Joseph's conjectures [17]. The interested reader might also want to consult [4] or [11].

Certain other variants of the commuting probability have also received attention. We take this opportunity to mention only a couple of those: Tărnăuceanu's [23] concept of the "subgroup permutability degree" as the probability that two subgroups of G permute. It turns out that this probability is an arithmetic measure of how close G is to being an Iwasawa group or, equivalently, a nilpotent modular group. A second interesting variant is due to Blackburn, Britnell, and Wildon. In [1], the authors introduce the probability that two elements, chosen independently and at random from a (finite) group G, are conjugate and study many of the fundamental properties that this probability enjoys.

We now focus our attention on the contents of the paper at hand. Let G be a finite group, and let H be a subgroup of G. By Hall's Marriage Theorem (see [13]), there exists a common set of representatives for the left and the right cosets of H in G; such a set is called a double or two-sided transversal of H in G. Even though the existence of such a double transversal is guaranteed, it seems unlikely that (in general) a left transversal of a non-normal subgroup will also be a right transversal. One of the main objectives of this paper is to compute the probability that a randomly chosen (with the uniform distribution) left transversal of H in G is also a right transversal. If the index of H in G is n and we denote by $\mathrm{DT}_G(H)$ the set of all double transversals of H in G, then the quotient

$$P_G(H) := \frac{|\mathrm{DT}_G(H)|}{|H|^n}$$

is precisely the probability in question. Actually, what we compute is the above probability in a more general setting, where two subgroups H, K of the same index are given, and we are seeking left transversals of H that are also right transversals for K; see Proposition 3.2.

In the present article, and in the spirit of $\mathrm{cp}(G)$, a novel invariant is introduced and studied: the transversal probability $\mathrm{tp}(G)$ of G defined as

$$\mathrm{tp}(G) := \min_{H \leqslant G} P_G(H).$$

Our objective here is two-fold: firstly, to compute $P_G(H)$ and to discuss its various numerical properties. In this direction, for example, we prove that if $n = (G : H)$, where H is not normal in G, then (see Corollary 3.10)

$$\frac{(n-1)!}{(n-1)^{n-1}} \leqslant P_G(H) \leqslant \frac{1}{2},$$

and (see Theorem 3.11) $\lim_{n\to\infty} P_G(H) = 0$; secondly, to investigate how $\mathrm{tp}(G)$ influences the structure of the group G. Results in this direction are for example the following.

- If $\mathrm{tp}(G) > (1/2)^{40}$, then either G is soluble or has a section isomorphic to A_5 (Proposition 4.3).
- If $\mathrm{tp}(G) > (1/2)^8$, then either G is supersoluble or has a section isomorphic to A_4 (Theorem 4.4).
- If $\mathrm{tp}(G) > (2/9)^2$, then either G is nilpotent or has a section isomorphic to one of the groups $\{A_4, D_3, D_5, D_7\}$ (Theorem 4.5).
- If $\mathrm{tp}(G) > 4/81$ and G is non-abelian, then it has derived length 2 (Proposition 4.14).

Furthermore, the results above are all sharp. In addition, the upper bound of $1/2$ obtained for $P_G(H)$ which holds for every non-normal $H \leqslant G$ carries over to $\mathrm{tp}(G)$ for every non-Dedekind group G, that is, $\mathrm{tp}(G) \leqslant 1/2$ for every such group. We are able to fully characterise groups G with $\mathrm{tp}(G) = 1/2$, as well as those with $\mathrm{tp}(G) = 1/4$ (cf. Corollary 4.11 and Theorem 4.13).

Finally, while investigating the possible values of $\mathrm{tp}(-)$, the following number theoretic result, of independent interest, was obtained (see Theorem 4.6).

- If
$$\prod_{i=1}^{n} \frac{t_i!}{t_i^{t_i}} = \prod_{i=1}^{k} \frac{p_i!}{p_i^{p_i}},$$
where the numbers $t_i > 1$ are positive integers and the p_i are distinct primes, then $k = n$ and, after appropriate rearrangement, $t_i = p_i$ for all $i = 1, \ldots, n$.

The article is organised as follows. In Section 2, the coset intersection graph is introduced and its properties needed for the computation of $P_G(-,-)$ are analysed. In Section 3, the function $P_G(-,-)$ is introduced and its precise value is calculated. A connection with the permanent of a matrix is established and several bounds on its values are given. In addition, the behaviour of $P_G(-)$ with respect to subgroups and quotients is determined. In Section 4, we work with the function $\mathrm{tp}(G)$, discussing the values it can and cannot attain, and the way it interacts with the structure of the group G. Finally, in Section 5, some open problems and questions are posed and discussed.

2 Coset intersection graphs

In this section, we introduce the concept of a coset intersection graph, which is the required background for the computation of $P_G(-,-)$. We follow the definition in [2].

Definition 2.1. Let G be a group, and let $H, K \leqslant G$ be subgroups. The *coset intersection graph* $\Gamma = \Gamma^G_{H,K}$ has vertex set $V(\Gamma) = G/H \sqcup K\backslash G$, two vertices being connected by an (undirected) edge if and only if the corresponding cosets have non-empty intersection. If $H = K$, we set $\Gamma^G_H := \Gamma^G_{H,H}$.

By definition, $\Gamma^G_{H,K}$ is a bipartite graph, split between $\{l_i H\}_{i\in I}$ and $\{Kr_j\}_{j\in J}$, where $\{l_i\}_{i\in I}$ is a left transversal for H in G and $\{r_j\}_{j\in J}$ is a right transversal for K in G. In this paper, G will always be a finite group; thus, the coset intersection graphs considered in what follows will all be finite. We also denote by $K_{a,b}$ the complete bipartite graph on a and b vertices, for any positive integers a and b.

Lemma 2.2. *Let $\Delta \leqslant \Gamma^G_{H,K}$ be a connected component. Then*

(i) $\bigcup_{gH\in V(\Delta)} gH = \bigcup_{Kg'\in V(\Delta)} Kg' = KgH$;

(ii) Δ *is a complete bipartite graph.*

Proof. (i) For given $g \in G$ with $gH \in V(\Delta)$, we have

$$gH \subseteq \bigcup_{\substack{Kg'\in K\backslash G \\ gH\cap Kg'\neq\emptyset}} Kg' \subseteq \bigcup_{Kg'\in V(\Delta)} Kg'$$

so that the left-hand side of (i) is contained in the right-hand side, and a similar argument establishes the reverse inclusion.

(ii) This is [2, Theorem 3]. □

Our next result, which (in somewhat different language and with a less direct proof) goes back to Ore [21], computes the number of connected components of a coset intersection graph.

Lemma 2.3. *The number of connected components of the graph* $\Gamma_{H,K}^G$ *equals the number* $|K\backslash G/H|$ *of* (K, H)*-double cosets of* G.

Proof. Denote by $\mathcal{C}(\Gamma)$ the collection of all connected components of $\Gamma = \Gamma_{H,K}^G$. If $aH, Kb \in V(\Delta)$ for some connected component Δ of Γ, then by Lemma 2.2 (ii), there exist elements $h \in H$ and $k \in K$ such that $ah = kb$ or $b = k^{-1}ah$. Hence, $KbH = KaH$. Keeping $aH \in V(\Delta)$ fixed while running through all right vertices Kb of Δ, we thus see that

$$KbH = KaH \quad (Kb \in V(\Delta)).$$

A symmetric argument, this time keeping a right vertex $Kb \in V(\Delta)$ fixed, and running through the left vertices aH of Δ shows that

$$KaH = KbH \quad (aH \in V(\Delta)).$$

It follows that the whole component Δ (meaning every vertex of Δ) is contained in one and the same (K, H)-double coset $Kg_\Delta H$, and sending Δ to $Kg_\Delta H$ yields a well-defined map

$$c\colon \mathcal{C}(\Gamma_{H,K}^G) \to K\backslash G/H, \quad c(\Delta) = KgH \quad (gH \in V(\Delta)).$$

Conversely, let $KgH \in K\backslash G/H$ be a given (K, H)-double coset. Then each left H-coset kgH and every right K-coset Kgh contained in KgH intersect non-trivially, as $kgh \in kgH \cap Kgh$. An argument analogous to the one given above now shows that all left H-cosets and all right K-cosets contained in KgH lie in one and the same connected component Δ_g of Γ, and sending KgH to Δ_g gives a well-defined map

$$d\colon K\backslash G/H \to \mathcal{C}(\Gamma_{H,K}^G), \quad d(KgH) = \Delta \quad (gH \in V(\Delta)).$$

The fact that $d(c(\Delta)) = \Delta$ for $\Delta \in \mathcal{C}(\Gamma)$ is now obvious (pin down Δ by means of a vertex $gH \in V(\Delta)$) and, similarly, we find that $c(d(KgH)) = KgH$, finishing the proof. □

With every edge of a coset intersection graph, we associate a weight in the following natural way.

Definition 2.4. Given a coset intersection graph $\Gamma = \Gamma_{H,K}^G$, we associate to an edge $e = l_i H - Kr_j$ in Γ the *weight* $w(e) := |l_i H \cap Kr_j|$.

If $g \in G$ and if $\Delta \leqslant \Gamma$ is a connected component of the coset intersection graph $\Gamma = \Gamma_{H,K}^G$, we shall write $g \in \Delta$ to mean $gH \in V(\Delta)$. Observe that, as $g \in gH \cap Kg$, we clearly have $gH \in V(\Delta)$ if and only if $Kg \in V(\Delta)$.

Lemma 2.5. *Let $\Delta \leqslant \Gamma_{H,K}^G$ be a connected component. Then all edges of Δ carry the same weight w, and we have*

$$w = |gHg^{-1} \cap K| = |H \cap g^{-1}Kg| \quad (g \in \Delta).$$

Proof. Let $aH, Kb \in V(\Delta)$ and $g \in aH \cap Kb$. Then $aH = gH$ and $Kb = Kg$, and thus

$$w_{a,b} := w(aH - Kb) = |aH \cap Kb| = |gH \cap Kg| = |gHg^{-1} \cap K|.$$

Keeping aH fixed, consider an arbitrary right coset $Kc \in V(\Delta)$, and set

$$w_{a,c} := w(aH - Kc).$$

Then, by the previous computation, $w_{a,c} = |xHx^{-1} \cap K|$ for any $x \in aH \cap Kc$. Since $x \in aH = gH$, we have $x = gh'$ for some $h' \in H$ so that

$$w_{a,c} = |gh'H(h')^{-1}g^{-1} \cap K| = |gHg^{-1} \cap K| = w_{a,b}.$$

Next, consider an arbitrary left coset $dH \in V(\Delta)$, noting that $dH - Kc$ is an arbitrary edge in Δ at this stage. Changing the roles of H and K in the previous argument, we also get $w_{a,c} = w_{d,c}$. Hence

$$w_{d,c} = w_{a,c} = w_{a,b} = w,$$

and our result follows. □

Definition 2.6. If $\Delta \leqslant \Gamma_{H,K}^G$ is a connected component, then we call the common weight of the edges in Δ the *weight* of Δ, denoted by $w(\Delta)$.

Lemma 2.7. *The following statements hold.*

(a) *If $\Delta_\sigma \cong K_{s_\sigma,t_\sigma}$ is a connected component of $\Gamma_{H,K}^G$, then*

$$|H| = w_\sigma t_\sigma \quad \text{and} \quad |K| = w_\sigma s_\sigma,$$

where $w_\sigma = w(\Delta_\sigma)$ is the weight of Δ_σ. In particular, $s_\sigma/t_\sigma = |K|/|H|$, and we have $s_\sigma = t_\sigma$ provided that $|H| = |K|$. In the latter case, the number m of components of type $K_{1,1}$ is given by

$$m = \begin{cases} (N_G(H) : H) & \text{if } H \text{ is conjugate to } K \text{ in } G, \\ 0 & \text{otherwise}, \end{cases} \tag{2.1}$$

and the weight of such a component equals $|H|$.

(b) *If* $G = \bigsqcup_{\sigma=1}^{s} Kg_\sigma H$, *then* $t_\sigma = (H : H \cap K^{g_\sigma})$, *where* $\Delta_\sigma \cong K_{t_\sigma,t_\sigma}$ *is the connected component of* $\Gamma_{H,K}^G$ *containing* g_σ.

Proof. (a) Let $\Delta_\sigma \cong K_{s_\sigma,t_\sigma}$ be a connected component of the coset intersection graph $\Gamma_{H,K}^G$ of weight $w(\Delta_\sigma) = w_\sigma$. By Lemma 2.5, each given left coset L_i of Δ_σ intersects every right coset $R_j \in V(\Delta_\sigma)$ in exactly w_σ elements and, as the right cosets of Δ_σ are pairwise disjoint, we have

$$\Big| L_i \cap \bigcup_{R_j \in V(\Delta_\sigma)} R_j \Big| = w_\sigma t_\sigma \quad (L_i \in V(\Delta_\sigma)).$$

However, $L_i \subseteq \bigcup_{R_j \in V(\Delta)} R_j$ by Lemma 2.2 (i) so that, in fact,

$$L_i \cap \bigcup_{R_j \in V(\Delta)} R_j = L_i,$$

and we find that

$$|H| = |L_i| = w_\sigma t_\sigma,$$

as claimed. A symmetric argument yields that $|K| = w_\sigma s_\sigma$. As concerns the assertion about the number m of trivial components, we note that $1 = t_\sigma = s_\sigma$ precisely when $w_\sigma = |H| = |K|$, which in turn implies (by Lemma 2.5) that

$$K = gHg^{-1} \cap K \quad \text{and} \quad H = H \cap g^{-1}Kg \quad \text{for } g \in \Delta_\sigma.$$

Hence, H, K are conjugate and

$$m = |\{g \in G : H^g = K\}|/|H|,$$

from which the given formula follows. The remaining assertions are now clear.

(b) This is a special case of [2, Proposition 6]. □

We record upper and lower bounds for the number $s = |K\backslash G/H|$ of connected components of a symmetric coset intersection graph $\Gamma_{H,K}^G$.

Lemma 2.8. *Let* $H, K \leqslant G$ *be such that*

$$(G : H) = n = (G : K),$$

and let $s = |K\backslash G/H|$. *Then we have*

$$\frac{n-m}{|H|} + m \leqslant s \leqslant \frac{n-m}{p} + m,$$

where p *is the smallest prime divisor of* $|H|$, *and* m *is given by* (2.1).

Proof. Let

$$G = \bigsqcup_{\sigma=1}^{s} Kg_\sigma H,$$

where $g_\sigma \in \Delta_\sigma$ for $1 \leqslant \sigma \leqslant s$. We observe that

$$|Kg_\sigma H| = \frac{|H| \cdot |K|}{|H \cap K^{g_\sigma}|} = \frac{|H|^2}{|H \cap K^{g_\sigma}|}$$

and that

$$|H| \leqslant \frac{|H|^2}{|H \cap K^{g_\sigma}|} \leqslant |H|^2.$$

Here, the left-hand side is assumed if and only if $g_\sigma \in \{g \in G : K^g = H\}$, while the right-hand side is assumed if and only if $H \cap K^{g_\sigma} = 1$. It follows that

$$\begin{aligned} n|H| = |G| = \sum_{\sigma=1}^{s} |Kg_\sigma H| &= m|H| + \sum_{\sigma=m+1}^{s} |Kg_\sigma H| \\ &\leqslant m|H| + (s-m)|H|^2. \end{aligned}$$

Dividing both sides by $|H|$ yields $n \leqslant m + (s-m)|H|$ or equivalently

$$\frac{n-m}{|H|} + m \leqslant s.$$

This proves the first half of the assertion. The upper bound for s is obtained in a similar way as follows:

$$\begin{aligned} n|H| = |G| = \sum_{\sigma=1}^{s} |Kg_\sigma H| &= m|H| + \sum_{\sigma=m+1}^{s} |H| \cdot (H : H \cap K^{g_\sigma}) \\ &\geqslant m|H| + (s-m)\,p|H|. \end{aligned}$$

Dividing both sides by $|H|$ yields $n \geqslant m + (s-m)\,p$, or equivalently

$$\frac{n-m}{p} + m \geqslant s,$$

completing the proof. □

With the help of Lemma 2.8, we can characterise groups G having a non-normal subgroup H satisfying $|H\backslash G/H| = 2$, as the next proposition shows.

Proposition 2.9. *Let G be a finite group and let $H \leqslant G$ be non-normal and of index n in G. If $s = |H\backslash G/H| = 2$ and $|H| = n - 1$, then G is a Frobenius group and H is a Frobenius complement. Conversely, suppose that G is a Frobenius group and H is a Frobenius complement. Then $s = 2$ if and only if $|H| = n - 1$.*

Proof. By Lemma 2.8 with $H = K$, we have

$$2 \geqslant \frac{n-m}{|H|} + m,$$

which implies that $m = 1$, since H is not normal in G (i.e., $m < n$); thus, we have $N_G(H) = H$. Now let $g \in G - H$. Since $s = 2$, we have $G = H \sqcup HgH$, and therefore

$$|G| = |H| + \frac{|H|^2}{|H \cap H^g|}.$$

Consequently, for $s = 2$ and $|H| = n - 1$,

$$n = 1 + \frac{|H|}{|H \cap H^g|} = 1 + \frac{n-1}{|H \cap H^g|}.$$

It follows that $|H \cap H^g| = 1$; thus, $H \cap H^g = 1$. Since g was arbitrary subject to lying outside H, we deduce that G is a Frobenius group with Frobenius complement H.

Suppose now that G is a Frobenius group and that H is a Frobenius complement. Then we have $s = 2$ if and only if $G = H \sqcup HgH$ for every $g \in G - H$ or, equivalently, if and only if

$$|G| = |H| + |H|^2,$$

that is, if and only if $|H| = n - 1$, which is as desired. □

3 The common transversal probability $P_G(-,-)$ of given subgroups

Let G be a finite group, and let $H \leqslant G$ be a subgroup. As we have noted before, a given left transversal for H in G may or may not be a two-sided one. The aim of this section is to compute the probability $P_G(H)$ that this happens. We actually give a generalised definition of the above probability, where two groups H, K of the same index n are concerned, and compute its value. This way we are able to show that the bigger the index n is, the smaller the probability of a left transversal of H being a right transversal of K is (see Theorem 3.11 below). In addition, we give bounds for the values of $P_G(H, K)$ and associate $P_G(H, K)$ with a permanent of a doubly stochastic matrix. Finally, in the last subsection, we show that $P_G(-)$ behaves well with respect to subgroups and homomorphic images.

3.1 Definition and computation of $P_G(-,-)$

We start with a definition of the main actors of this paper.

Definition 3.1. Let G be a finite group, and let $H, K \leqslant G$ be subgroups such that $(G : H) = n = (G : K)$.

(i) We denote by $\mathrm{DT}_G(H, K)$ the set of all left transversals for H in G, which are also right transversals for K in G.

(ii) We let

$$P_G(H, K) := \frac{|\mathrm{DT}_G(H, K)|}{|H|^n}$$

so that $P_G(H, K)$ is the (Laplacian) probability that a left transversal for H in G is also a right transversal for K in G. If $H = K$, we let

$$P_G(H) := P_G(H, H)$$

be the probability that a left transversal for H in G is a two-sided one. We call $P_G(H, K)$ the *common transversal probability* for the subgroups $H, K \leqslant G$.

(iii) We set

$$\mathrm{tp}(G) := \min_{H \leqslant G} P_G(H),$$

which is an invariant of G alone, termed the *common transversal probability* of the group G.

The first result of this section computes the probability $P_G(H, K)$ in terms of the sizes of the connected components of the coset intersection graph $\Gamma^G_{H,K}$.

Proposition 3.2. *Let G be a finite group, and let $H, K \leqslant G$ be subgroups such that $(G : H) = (G : K) = n$. Assume further that $|K\backslash G/H| = s$ and let $\Delta_\sigma = K_{t_\sigma, t_\sigma}$ for $\sigma \in [s]$ denote the connected components of $\Gamma = \Gamma^G_{H,K}$. Then*

$$P_G(H, K) = \prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}}, \tag{3.1}$$

where $t_\sigma = |Kg_\sigma H|/|K|$ with $Kg_\sigma H$ the double coset corresponding to σ. So $t_\sigma \mid |H|$, and $\sum_{\sigma=1}^{s} t_\sigma = n$.

Proof. It is clear from the definition of Γ that the number of common transversals for the pair (H, K) in G equals the product over the number of common transversals for the t_σ left-H and right-K cosets in each component Δ_σ of Γ. Moreover,

the fact that each component Δ_σ is a symmetric complete bipartite graph K_{t_σ,t_σ}, while the weight w_σ on every edge of Δ_σ is the size of the intersection of any left-H with any right-K coset of Δ_σ, implies that the number of common (H, K)-double transversals for the t_σ left-H and t_σ right-K cosets of Δ_σ equals $t_\sigma! \cdot w_\sigma^{t_\sigma}$. By Lemma 2.7, we have $w_\sigma = |H|/t_\sigma$; in particular, $t_\sigma \mid |H|$ for each $\sigma \in [s]$. Since $\sum_\sigma t_\sigma = n$, it follows that

$$\begin{aligned} |\mathrm{DT}_G(H,K)| &= \prod_{\sigma=1}^{s} t_\sigma!\, w_\sigma^{t_\sigma} = \prod_{\sigma=1}^{s} \Big(\frac{t_\sigma!}{t_\sigma^{t_\sigma}} \cdot |H|^{t_\sigma} \Big) \\ &= |H|^{\sum_\sigma t_\sigma} \cdot \prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} = |H|^n \cdot \prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}}. \end{aligned}$$

Hence,

$$P_G(H,K) = \frac{|\mathrm{DT}_G(H,K)|}{|H|^n} = \prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}},$$

as claimed. □

Definition 3.3. Let $H \leqslant G$ and let $\{\Delta_{t_\sigma}\}_\sigma$ be the family of connected components of Γ_H^G, where the family of positive integers $\{t_\sigma\}$ is defined as above. We list the integers t_σ in decreasing order, say $(t_1, \ldots, t_s)$, and call the resulting vector the t-vector of H in G.

Corollary 3.4. *We have $P_G(H, K) = 1$ if and only if $H = K \trianglelefteq G$. Hence, we have* $\mathrm{tp}(G) = 1$ *if and only if G is a Dedekind group, that is, if and only if every subgroup in G is normal.*

Proof. This follows immediately from Proposition 3.2. □

Corollary 3.5. *Suppose that $H, K \leqslant G$ are such that $(G : H) = n = (G : K)$, and that $H \cap K^g = 1$ for $g \notin KH$. Then*

$$P_G(H,K) = \Big(\frac{|H|!}{|H|^{|H|}} \Big)^{\frac{n-m}{|H|}}.$$

Proof. Apart from the double coset HK for every other $g_\sigma \in \Delta_\sigma$, we have

$$t_\sigma = \frac{|H|}{|H \cap K^{g_\sigma}|} = |H|, \quad g_\sigma \notin KH,$$

by Lemma 2.7 (b). Hence

$$P_G(H,K) = \Big(\frac{|H|!}{|H|^{|H|}} \Big)^{s-1} = \Big(\frac{|H|!}{|H|^{|H|}} \Big)^{\frac{n-1}{|H|}}$$

by Proposition 3.2, plus the fact that $n = \sum_{\sigma=1}^{s} t_\sigma = 1 + |H|(s-1)$. □

Corollary 3.6. *Let G be a finite Frobenius group with Frobenius complement H, and let $(G : H) = n$. Then*

$$P_G(H) = \left(\frac{|H|!}{|H|^{|H|}}\right)^{\frac{n-1}{|H|}}.$$

Proof. Here $H = K$, $H \cap H^g = 1$ for $g \notin H$, and $N_G(H) = H$, whence the result by Corollary 3.5. □

Corollary 3.7. *Let $H \leqslant G$ be a subgroup of order $|H| = p$ a prime, and suppose that $(G : H) = n$. Then*

$$P_G(H) = \left(\frac{p!}{p^p}\right)^{\frac{n-m}{p}},$$

where $m = (N_G(H) : H)$, and the exponent on the right-hand side is an integer.

Proof. Apart from m trivial components in $\Gamma_H^G := \Gamma_{H,H}^G$, we have $\frac{n-m}{p}$ components $\Delta_\sigma \cong K_{p,p}$. In particular, this quotient is an integer. The result now follows from Proposition 3.2. □

3.2 The function P_G and permanents

At this point, we wish to briefly discuss a connection between the probability $P_G(H, K)$, where H and K are subgroups of the finite group G such that

$$(G : H) = n = (G : K),$$

and the permanent of a certain associated matrix (for properties of permanents used in what follows, we refer to the standard reference [20]). The matrix we associate to the triple (G, H, K) is the $n \times n$ weight matrix $W = W_{H,K}^G = (w_{i,j})$, whose (i, j) entry is $w_{i,j} = |l_i H \cap K r_j|$. That is, $w_{i,j}$ is the weight assigned to the edge $e = l_i H - K r_j$ of the coset intersection graph $\Gamma = \Gamma_{H,K}^G$. A simple combinatorial argument shows that

$$|\mathrm{DT}_G(H, K)| = \sum_{\sigma \in S_n} \prod_{i=1}^{n} w_{i,\sigma(i)}.$$

As the right-hand side in the above equation is exactly the permanent of the matrix W, we get $|\mathrm{DT}_G(H, K)| = \mathrm{per}(W_{H,K}^G)$. Observe now that all row and column sums of W are equal to $|H| = |K|$, that is,

$$\sum_{i=1}^{n} w_{i,r} = |H| = \sum_{j=1}^{n} w_{t,j} \quad (1 \leqslant r, t \leqslant n).$$

Thus, the matrix

$$M = M_{H,K}^G := \frac{1}{|H|} \cdot W_{H,K}^G$$

is an $n \times n$ doubly stochastic matrix (all row and column sums equal to 1); furthermore, its permanent is exactly the common transversal probability for the subgroups $H, K \leqslant G$, as

$$\mathrm{per}(M) = \frac{1}{|H|^n} \cdot \mathrm{per}(W_{H,K}^G) = P_G(H, K).$$

In addition, after appropriate row and column permutations, we can group together cosets (left for H and right for K) which intersect. In this way, M is seen to be equivalent to a block diagonal matrix: there exist $n \times n$ permutation matrices P and Q such that

$$PMQ = \begin{bmatrix} M_1 & 0 & \cdots & 0 \\ 0 & M_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & M_s \end{bmatrix},$$

where, for $\sigma = 1, \ldots, s$, the matrix M_σ is the $t_\sigma \times t_\sigma$ rational matrix whose entries all equal $\frac{1}{t_\sigma}$. As usual, $t_\sigma = (H : H \cap K^{g_\sigma})$, where $\{t_\sigma\}_{\sigma=1}^s$ is the family of sizes of the connected components of the coset intersection graph $\Gamma_{H,K}^G$. Collecting together these observations, we get most of the following.

Proposition 3.8. *The doubly stochastic matrix $M = M_{H,K}^G$ satisfies*

$$\mathrm{per}(M) = P_G(H, K) = \prod_{\sigma=1}^{s} \frac{(t_\sigma)!}{t_\sigma^{t_\sigma}}. \tag{3.2}$$

The eigenvalues of M are 0 with multiplicity $n - s$, and 1 with multiplicity s. Moreover, M is positive semi-definite.

Proof. We have already seen that $\mathrm{per}(M) = P_G(H, K)$ and the right-hand side equals the desired product according to Proposition 3.2. We remark here that an alternative way to see that

$$\mathrm{per}(M_{H,K}^G) = \prod_{\sigma=1}^{s} \frac{(t_\sigma)!}{t_\sigma^{t_\sigma}}$$

is the following: since $PMQ = \mathrm{diag}(M_1, \ldots, M_s)$, we have

$$\mathrm{per}(PMQ) = \mathrm{per}(M) = \prod_{\sigma=1}^{s} \mathrm{per}(M_\sigma).$$

On the other hand, working directly from the definition, the permanent of a square matrix $M = (m_{i,j})$ of size c with all entries equal to some fixed number a is seen to be

$$\mathrm{per}(M) = \sum_{\sigma \in S_c} \prod_{i=1}^{c} m_{i,\sigma(i)} = \sum_{\sigma \in S_c} a^c = c!\, a^c.$$

Specifying $a = 1/t_\sigma$ and $c = t_\sigma$, for each $\sigma \in [s]$, yields

$$\mathrm{per}(M_\sigma) = t_\sigma!/t_\sigma^{t_\sigma},$$

whence (3.2). The assertion concerning the eigenvalues of M follows from the fact that the characteristic polynomial of the matrix M_σ is

$$\det(M_\sigma - \lambda I) = (-\lambda)^{t_\sigma - 1}(1 - \lambda),$$

which implies that the characteristic polynomial of the matrix M is given by

$$\det(M - \lambda I) = (-\lambda)^{n-s}(1 - \lambda)^s.$$

Thus, $\mathrm{tr}(M) = s$, while $\det(M) = 0$, which can, of course, also be seen directly. Also, $\mathrm{rk}(M) = s$. Since M is symmetric and its eigenvalues are non-negative, it follows that M is positive semi-definite. □

3.3 A limit theorem

The aim in this subsection is twofold. Firstly, we will give sharp bounds for the value $P_G(H, K)$, so, for example, we will prove that, for non-normal conjugate subgroups, its value cannot exceed $1/2$. Secondly, we will show that the bigger the index $n = (G : H) = (G : K)$ is, the smaller the value $P_G(H, K)$ becomes. We start with the following purely arithmetic result.

Lemma 3.9. *Let $s \geqslant 1$, and let $\{t_\sigma\}_{\sigma=1}^s$ be a family of positive integers with*

$$\sum_{\sigma=1}^{s} t_\sigma = n.$$

Then

$$\frac{n!}{n^n} \leqslant \prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} \leqslant \Big(\frac{n+s}{2n}\Big)^n. \tag{3.3}$$

Furthermore, if m of the t_σ are equal to 1, then

$$\frac{(n-m)!}{(n-m)^{n-m}} \leqslant \prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} \leqslant \frac{1}{2^{s-m}}. \tag{3.4}$$

Proof. We first prove the left-hand inequality in (3.3) by induction on s. For $s = 1$, both sides are equal. Next, suppose that $s = 2$ so that $t_1 + t_2 = n$. Then $t_1! \cdot t_2! = \binom{n}{t_1}^{-1} n!$, while $n^n = (t_1 + t_2)^n \geqslant \binom{n}{t_1} t_1^{t_1} \cdot t_2^{t_2}$. Thus,

$$\frac{n!}{n^n} = \frac{\binom{n}{t_1} t_1! \cdot t_2!}{n^n} \leqslant \frac{\binom{n}{t_1} t_1! \cdot t_2!}{\binom{n}{t_1} t_1^{t_1} \cdot t_2^{t_2}} = \prod_{\sigma=1}^{2} \frac{t_\sigma!}{t_\sigma^{t_\sigma}},$$

as desired. For the induction step, suppose that the left-hand inequality in (3.3) holds for all $s \leqslant u$ with some $u \geqslant 2$, and let $n = t_1 + t_2 + \cdots + t_{u+1}$. Then we have $n = t_1 + (n - t_1)$, and the inductive hypothesis for $s = 2$ and $s = u$ gives

$$\frac{n!}{n^n} \leqslant \frac{t_1!}{t_1^{t_1}} \cdot \frac{(n - t_1)!}{(n - t_1)^{n - t_1}} \leqslant \frac{t_1!}{t_1^{t_1}} \cdot \prod_{\sigma=2}^{u+1} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} = \prod_{\sigma=1}^{u+1} \frac{t_\sigma!}{t_\sigma^{t_\sigma}},$$

whence our result.

The right-hand inequality in (3.3) follows from the arithmetic-geometric mean inequality as follows:

$$\begin{aligned}
\prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} = \prod_{\sigma=1}^{s} \prod_{\tau=1}^{t_\sigma} \tau / t_\sigma &\leqslant \left(\sum_{\sigma=1}^{s} \sum_{\tau=1}^{t_\sigma} \tau / t_\sigma \Big/ \sum_{\sigma=1}^{s} t_\sigma \right)^{\sum_{\sigma=1}^{s} t_\sigma} \\
&= \left(\frac{1}{n} \sum_{\sigma=1}^{s} \frac{1}{t_\sigma} \frac{t_\sigma (t_\sigma + 1)}{2} \right)^n \\
&= \left(\sum_{\sigma=1}^{s} \frac{t_\sigma + 1}{2n} \right)^n = \left(\frac{n + s}{2n} \right)^n.
\end{aligned}$$

This establishes the first part of the lemma. Now suppose that

$$t_1 = t_2 = \cdots = t_m = 1$$

for some $m \leqslant s$, while $t_\sigma \geqslant 2$ for $m + 1 \leqslant \sigma \leqslant s$. Then $n - m = t_{m+1} + \cdots + t_s$; thus, by the left-hand inequality in (3.3),

$$\frac{(n - m)!}{(n - m)^{n-m}} \leqslant \prod_{\sigma=m+1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} = \prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}}.$$

Furthermore, for each $\sigma > m$, we have $t_\sigma \geqslant 2$ so that $t_\sigma! / t_\sigma^{t_\sigma} \leqslant 1/2$; thus

$$\prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} \leqslant 2^{-(s-m)}.$$

This completes the proof of the lemma. □

Corollary 3.10. *Let G be a finite group, and let $H, K \leqslant G$ be conjugate, non-normal subgroups of index n in G. Then we have*

$$\frac{(n-1)!}{(n-1)^{n-1}} \leqslant P_G(H,K) \leqslant \frac{1}{2}, \tag{3.5}$$

and these bounds are best possible. In particular, $P_G(H,K) = 1/2$ for $n = 3$.

Proof. Let H, K be conjugate subgroups of G. Then, according to Lemma 2.7 (a), the number m of connected components Δ_σ of $\Gamma^G_{H,K}$ of type $\Delta_\sigma \cong K_{1,1}$ is at least one, i.e., $m \geqslant 1$. On the other hand, we cannot have $m = s = |K\backslash G/H|$, since otherwise $P_G(H,K) = 1$ by Proposition 3.2, so that $H = K \trianglelefteq G$ by Corollary 3.4, contradicting our hypothesis. Hence, we have $1 \leqslant m < s$. Since the left-hand side of (3.4) decreases with m, while the right-hand side increases with m, the result follows.

We next focus on the sharpness of the bounds in (3.5). Let F be a field with q elements, where q is a power of a prime. Then the action of the multiplicative group F^* on the additive group of F is of Frobenius type. The resulting group $G = F \rtimes F^\times$ is a Frobenius group with Frobenius complement $H \cong F^*$ of order $q-1$ and index q. Thus, by Corollary 3.6,

$$P_G(H) = \left[\frac{(q-1)!}{(q-1)^{q-1}}\right]^{\frac{q-1}{q-1}} = \frac{(q-1)!}{(q-1)^{q-1}},$$

which shows that the lower bound in equation (3.5) is attained.

Also, for $q = 3$, let $G \cong S_3$, and let H be any Sylow 2-subgroup of G. In this case, $P_G(H) = 1/2$ so that the upper bound in (3.5) is attained as well. □

We can now establish our first main result.

Theorem 3.11. *Let G be a finite group, and let $H, K \leqslant G$ be subgroups of the same index n such that at least one of them is not normal in G. Then*

$$P_G(H,K) \leqslant \left(\frac{7}{8}\right)^n.$$

In particular, $\lim_{n\to\infty} P_G(H) = 0$.

Proof. According to Proposition 3.2 and Lemma 3.9, we have

$$P_G(H,K) = \prod_{i=1}^{s} \frac{t_i!}{t_i^{t_i}} \leqslant \left(\frac{n+s}{2n}\right)^n, \tag{3.6}$$

where $s = |K\backslash G/H|$ equals the number of connected components of $\Gamma^G_{H,K}$. According to Lemma 2.7, exactly m of those t_σ's are equal to 1, while for the rest, we get $t_\sigma \geqslant 2$, where $m = (N_G(H) : H)$ if H, K are conjugate, or $m = 0$ if they

are not. As $\sum_{\sigma=1}^{s} t_\sigma = n$, we conclude that

$$s \leqslant m + \frac{n-m}{2} = \frac{n+m}{2}.$$

Furthermore, by (2.1), m is either zero, or divides n *properly* (otherwise we would have $H = K \trianglelefteq G$ contradicting our hypothesis); thus, $m \leqslant \frac{n}{2}$. Hence, $s \leqslant \frac{3n}{4}$ in either case, and equation (3.6) implies

$$P_G(H,K) \leqslant \left(\frac{n+s}{2n}\right)^n \leqslant \left(\frac{7}{8}\right)^n, \tag{3.7}$$

completing the proof. □

3.4 An improved bound for $P_G(H,K)$

We digress briefly to discuss an upper bound for the function $\prod_{\sigma=1}^{s} t_\sigma!/t_\sigma^{t_\sigma}$ which is stronger than (3.7). For $x > 0$, let

$$f(x) := \frac{\Gamma(x+1)}{x^x}$$

where

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt$$

is the well-known Γ-function. Observe that $f(t_\sigma) = t_\sigma!/t_\sigma^{t_\sigma}$ and thus

$$\prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} = \prod_{\sigma=1}^{s} f(t_\sigma).$$

We argue that f is strictly log-concave, that is, the function $g(x) = \log f(x)$ is strictly concave. First, g is infinitely differentiable. Second, we have

$$g''(x) = \psi'(x+1) - \frac{1}{x},$$

where $\psi(x) = (\log \Gamma(x))' = \Gamma'(x)/\Gamma(x)$ is the so-called digamma function. By [9, Lemma 2], we have

$$\psi'(x+1) < e^{\frac{1}{x+1}} - 1 \quad (x > 0),$$

while

$$e^{\frac{1}{x+1}} - 1 < \frac{1}{x} \quad (x > 0)$$

by virtue of the elementary inequality

$$e < \left(1 + \frac{1}{x}\right)^{x+1},$$

which is valid for all $x > 0$. Thus

$$g''(x) = \psi'(x+1) - \frac{1}{x} < 0 \quad (x > 0),$$

which implies that g is strictly concave as claimed.

Now let

$$\Pi := \prod_{\sigma=1}^{s} f(t_\sigma).$$

Then

$$\log \Pi = \sum_{\sigma=1}^{s} \log f(t_\sigma) \leqslant s \cdot \log f\left(\frac{\sum_{\sigma=1}^{s} t_\sigma}{s}\right) = s \cdot \log f\left(\frac{n}{s}\right),$$

where the inequality is a consequence of Jensen's inequality applied to the concave function $g(x) = \log f(x)$. We conclude that

$$\prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} = \prod_{\sigma=1}^{s} f(t_\sigma) \leqslant f\left(\frac{n}{s}\right)^s.$$

At this point, we are interested in comparing the two bounds

$$f\left(\frac{n}{s}\right)^s \quad \text{and} \quad \left(\frac{n+s}{2n}\right)^n.$$

Specifically, we will show the following.

Proposition 3.12. *For all $n \in \mathbb{N}$ and for all s with $s \leqslant \frac{3}{4}n$, we have*

$$f\left(\frac{n}{s}\right)^s \leqslant c^n \left(\frac{n+s}{2n}\right)^n, \tag{3.8}$$

where

$$c := \frac{8}{7} f\left(\frac{4}{3}\right)^{3/4} = 0.976986\ldots.$$

Proof. Proving inequality (3.8) is tantamount to showing that $f(y) \leqslant c^y \left(\frac{y+1}{2y}\right)^y$ for all $y \geqslant 4/3$, where we have made the substitution $y := n/s$. It will clearly suffice to prove that $\log f(y) \leqslant y\left[\log\left(\frac{y+1}{2y}\right) + \log c\right]$. So let

$$\begin{aligned} H(y) &:= \log f(y) - y\left[\log\left(\frac{y+1}{2y}\right) + \log c\right] \\ &= \log \Gamma(y+1) - y \log y - y\left[\log\left(\frac{y+1}{2y}\right) + \log c\right] \\ &= \log \Gamma(y+1) - y\left[\log\left(\frac{y+1}{2}\right) + \log c\right] \end{aligned}$$

for $y \geqslant 4/3$, and observe that $H(4/3) = 0$. Thus, the claim will have been established, provided that we can show that $H'(y) \leqslant 0$ for $y \geqslant 4/3$. Now observe that

$$H'(y) = \psi(y+1) - \log\Big(\frac{y+1}{2}\Big) - \log c - \frac{y}{y+1}.$$

By [9, Lemma 1], we have

$$\psi(y+1) < \log(y+1) - \frac{1}{2(y+1)} \quad (y > 0).$$

Hence,

$$\begin{aligned} H'(y) &< \log(y+1) - \frac{1}{2(y+1)} - \log\Big(\frac{y+1}{2}\Big) - \log c - \frac{y}{y+1} \\ &= \frac{y[2\log(2/c) - 2] + 2\log(2/c) - 1}{2(y+1)}. \end{aligned}$$

We note that the numerator N in the last expression is negative since $N \geqslant 0$ would imply that

$$y \leqslant \frac{2\log(2/c) - 1}{2 - 2\log(2/c)} \approx 0.763233 < \frac{4}{3}$$

contradicting our assumption that $y \geqslant 4/3$, while the denominator is positive. Thus, we have $H'(y) \leqslant 0$, which is what we wanted to prove. □

Remark 3.13. It is clear that we have equality in (3.8) if and only if $n = (4/3)s$ since $H(4/3) = 0$ and $H'(y)$ is in fact strictly negative for $y > 4/3$. Moreover, if s remains bounded, and thus $y = n/s$ grows without bound, $f(n/s)^s$ becomes an even better bound than $\big(\frac{n+s}{2n}\big)^n$ since

$$f\Big(\frac{n}{s}\Big)^s \sim \frac{(2\pi\frac{n}{s})^{s/2}}{e^n}, \quad \text{while} \quad \Big(\frac{n+s}{2n}\Big)^n \sim \frac{e^s}{2^n},$$

where the first relation follows from Stirling's approximation.

3.5 Subgroups, homomorphic images, and the function $P_G(-)$

From now on, we shall concentrate on the case where $H = K$, that is, we shall focus on the functions of one variable $P_G(-)$ and $\mathrm{tp}(-)$. Our aim in this subsection is to show that $P_G(-)$ behaves well with respect to subgroups and homomorphic images.

Proposition 3.14. *Let $K \leqslant H \leqslant G$ be finite groups. Then every connected component of Γ_K^H is also a connected component of Γ_K^G; in particular, Γ_K^H is an induced subgraph of Γ_K^G, and we have*

$$P_G(K) \leqslant P_H(K). \tag{3.9}$$

Proof. First, it is clear that Γ_K^H is a subgraph of Γ_K^G. Second, we observe that

$$hK \cap Kg \neq \emptyset \implies g \in H \quad (h \in H,\, g \in G)$$

since, by hypothesis, there have to exist elements $k_1, k_2 \in K$ such that $hk_1 = k_2 g$, so that $g = k_2^{-1} h k_1 \in H$, as $K \leqslant H$. Therefore, an edge in Γ_K^G with one vertex in the set H/K necessarily has its other bounding vertex in $K\backslash H$, which implies that the connected components of Γ_K^H are *identical* with certain components of Γ_K^G. Third, we obviously have

$$s_H := |K\backslash H/K| \leqslant |K\backslash G/K| =: s_G.$$

Thinking of the connected components of Γ_K^H as being listed first among the components of Γ_K^G, we now find that

$$P_G(K) = \prod_{\sigma=1}^{s_G} \frac{(t_\sigma)!}{t_\sigma^{t_\sigma}} = \prod_{\sigma=1}^{s_H} \frac{(t_\sigma)!}{t_\sigma^{t_\sigma}} \times \prod_{\sigma=s_H+1}^{s_G} \frac{(t_\sigma)!}{t_\sigma^{t_\sigma}} \leqslant \prod_{\sigma=1}^{s_H} \frac{(t_\sigma)!}{t_\sigma^{t_\sigma}} = P_H(K),$$

whence (3.9). □

Our next result concerns the connection of P_G with homomorphic images.

Theorem 3.15. *Let G and K be finite groups, $f: G \to K$ a group homomorphism, and let $H \leqslant G$ be a subgroup. Then, if $N := \ker(f) \leqslant H$, we have*

$$P_G(H) = P_{f(G)}(f(H)).$$

Proof. Our aim is to define and analyse a certain map

$$\Phi: \mathrm{DT}_G(H) := \mathrm{DT}_G(H, H) \to \mathrm{DT}_{f(G)}(f(H)).$$

Suppose that $(G : H) = n$, and let $T = \{t_1, t_2, \ldots, t_n\}$ be a two-sided transversal for H in G. Applying the homomorphism f, we obtain what looks at first like a multiset $\overline{T} = \{f(t_1), f(t_2), \ldots, f(t_n)\} \subseteq K$, and we claim that $\overline{T}$ is in fact a two-sided transversal for $f(H)$ in $f(G)$; in particular, $\overline{T}$ is an n-set. Suppose first that

$$f(t_{\nu_1} H) \cap f(t_{\nu_2} H) = f(t_{\nu_1}) f(H) \cap f(t_{\nu_2}) f(H) \neq \emptyset \quad (\nu_1, \nu_2 \in [n]).$$

Then there exist elements $h_1, h_2 \in H$ such that

$$f(t_{\nu_1}h_1) = f(t_{\nu_2}h_2) \quad \text{or} \quad f(t_{\nu_1}h_1h_2^{-1}t_{\nu_2}^{-1}) = 1.$$

Thus, there are elements $n, n' \in N$ such that

$$t_{\nu_2} = n^{-1}t_{\nu_1}h_1h_2^{-1} = t_{\nu_1}n'h_1h_2^{-1} \in t_{\nu_1}H,$$

as $N \leqslant H$ by hypothesis. This forces $\nu_1 = \nu_2$ since T is a left transversal for H in G. Since $(f(G) : f(H)) \leqslant n$, it follows that $\overline{T}$ is a complete set of pairwise inequivalent representatives for the left cosets of $f(H)$ in $f(G)$. Similarly, if

$$f(Ht_{\nu_1}) \cap f(Ht_{\nu_2}) = f(H)f(t_{\nu_1}) \cap f(H)f(t_{\nu_2}) \neq \emptyset \quad (\nu_1, \nu_2 \in [n]),$$

we find that $f(h_1t_{\nu_1}t_{\nu_2}^{-1}h_2^{-1}) = 1$ for some $h_1, h_2 \in H$, or

$$t_{\nu_1}t_{\nu_2}^{-1} = h_1^{-1}nh_2 \in H \quad (n \in N),$$

again implying $\nu_1 = \nu_2$, so that $\overline{T}$ is also a right transversal for $f(H)$ in $f(G)$; therefore, $\overline{T} \in \mathrm{DT}_{f(G)}(f(H))$.

To obtain a well-defined map from $\mathrm{DT}_G(H)$ to $\mathrm{DT}_{f(G)}(f(H))$, we now set $\Phi(T) := \overline{T}$ and claim that Φ is surjective. In order to justify this claim, let

$$\{f(s_\nu)\}_{1\leqslant\nu\leqslant n} \in \mathrm{DT}_{f(G)}(f(H)),$$

pick arbitrary pre-images $\tilde{s}_\nu \in f^{-1}(f(s_\nu))$ for $1 \leqslant \nu \leqslant n$, and form the set

$$\tilde{S} = \{\tilde{s}_\nu : 1 \leqslant \nu \leqslant n\}.$$

Then $\tilde{S} \in \mathrm{DT}_G(H)$. Indeed, if $\tilde{s}_{\nu_1}^{-1}\tilde{s}_{\nu_2} \in H$, then

$$f(\tilde{s}_{\nu_1}^{-1}\tilde{s}_{\nu_2}) = f(\tilde{s}_{\nu_1})^{-1}f(\tilde{s}_{\nu_2}) = f(s_{\nu_1})^{-1}f(s_{\nu_2}) \in f(H)$$

so that $\nu_1 = \nu_2$. A similar argument works for right cosets and, by construction,

$$\Phi(\tilde{S}) = \{f(\tilde{s}_\nu)\}_{1\leqslant\nu\leqslant n} = \{f(s_\nu)\}_{1\leqslant\nu\leqslant n},$$

whence our claim.

We observe that, as $N = \ker(f) \leqslant H$, if $T = \{t_\nu\}_{1\leqslant\nu\leqslant n} \in \mathrm{DT}_G(H)$, then so is $T' = \{m_\nu t_\nu\}_{1\leqslant\nu\leqslant n}$, where $m_1, \ldots, m_n \in N$, as

$$Hm_\nu t_\nu = Ht_\nu \quad \text{and} \quad m_\nu t_\nu H = t_\nu m'_\nu H = t_\nu H,$$

where $m'_\nu \in N$. Moreover, we have $\Phi(T) = \Phi(T')$ since $f(m_\nu t_\nu) = f(t_\nu)$. Consequently, for each $\overline{T} \in \mathrm{DT}_{f(G)}(f(H))$, there are at least $|N|^n$ distinct pre-images under Φ. Actually, we have

$$|\Phi^{-1}(\overline{T})| = |N|^n \quad (\overline{T} \in \mathrm{DT}_{f(G)}(f(H))).$$

Namely, if $S = \{s_\nu\}_{1 \leqslant \nu \leqslant n} \in \mathrm{DT}_G(H)$ with $\Phi(S) = \Phi(T)$, where $T \in \mathrm{DT}_G(H)$ is a given two-sided transversal, then after appropriate rearrangement of indices, we have $f(s_\nu) = f(t_\nu)$ for $1 \leqslant \nu \leqslant n$; thus,

$$s_\nu = m_\nu t_\nu \quad (1 \leqslant \nu \leqslant n),$$

where $m_\nu \in N$, as claimed. Summarising our findings so far, we have shown that

$$|\mathrm{DT}_{f(G)}(f(H))| = \frac{|\mathrm{DT}_G(H)|}{|N|^n}.$$

It follows that

$$\begin{aligned} P_{f(G)}(f(H)) = \frac{|\mathrm{DT}_{f(G)}(f(H))|}{|f(H)|^n} &= \frac{|\mathrm{DT}_G(H)|/|N|^n}{|H/N|^n} \\ &= \frac{|\mathrm{DT}_G(H)|}{|H|^n} = P_G(H), \end{aligned}$$

completing the proof. □

Corollary 3.16. *Let G be a finite group, H a subgroup of G, and let $f: G \to K$ be an isomorphism. Then $P_G(H) = P_K(f(H))$; in particular,* $\mathrm{tp}(G) = \mathrm{tp}(K)$*, that is,* $\mathrm{tp}(-)$ *is an isomorphism invariant.*

Taking $f: G \to G/N$ the canonical projection, where N is a normal subgroup of G, Theorem 3.15 gives the following.

Corollary 3.17. *Let G be a finite group, and let $N \leqslant H$ be subgroups of G with N normal in G. Then $P_G(H) = P_{G/N}(H/N)$.*

4 The common transversal probability tp(−) of a group

We now turn our attention to the invariant $\mathrm{tp}(-)$, defined as

$$\mathrm{tp}(G) := \min_{H \leqslant G} P_G(H).$$

The main objective of the section is to relate the values of $\mathrm{tp}(G)$ to key properties of the group G. Roughly, this means that the larger $\mathrm{tp}(G)$ is, the more normal structure G exhibits; in particular, we show that solubility, supersolubility, and

nilpotency are characterised by certain values of $\mathrm{tp}(G)$. We also compute the common transversal probability of certain groups, and show that specific values of tp are achieved by specific groups, while others are not attained at all. We also show that tp behaves well with respect to subgroups, quotients, and group extensions.

4.1 The function tp(−) and group structure

Our first result records the behaviour of $\mathrm{tp}(-)$ under taking subgroups, quotients, and sections.

Proposition 4.1. *Let G be a finite group.*

(i) *For every proper subgroup H of G, we have* $\mathrm{tp}(G) \leqslant \mathrm{tp}(H)$.

(ii) *For every normal subgroup N of G, we have* $\mathrm{tp}(G) \leqslant \mathrm{tp}(G/N)$.

(iii) *For every section X of G, we have* $\mathrm{tp}(G) \leqslant \mathrm{tp}(X)$.

(iv) *If G is a non-Dedekind p-group, then* $\mathrm{tp}(G) \leqslant p!/p^p$.

Proof. (i) Let $K \leqslant H$ be such that $\mathrm{tp}(H) = P_H(K)$. Then, according to Proposition 3.14, we have $P_G(K) \leqslant P_H(K)$. Hence,

$$\mathrm{tp}(G) \leqslant P_G(K) \leqslant P_H(K) = \mathrm{tp}(H).$$

(ii) In view of Corollary 3.17, we have

$$\mathrm{tp}(G/N) = \min_{N \leqslant H \leqslant G} P_{G/N}(H/N) = \min_{N \leqslant H \leqslant G} P_G(H) \geqslant \mathrm{tp}(G).$$

(iii) Let $X = H/N$, where $N \trianglelefteq H \leqslant G$. Then, by parts (i) and (ii),

$$\mathrm{tp}(G) \leqslant \mathrm{tp}(H) \leqslant \mathrm{tp}(H/N) = \mathrm{tp}(X),$$

whence our result.

(iv) Suppose that $H \leqslant G$ is not normal in G and that $(G : H) = n$. Then the number m of trivial connected components of the graph Γ_H^G satisfies

$$m = (N_G(H) : H) < n.$$

Thus, there exists at least one non-trivial component $\Delta_\sigma \cong K_{t_\sigma, t_\sigma}$ in Γ_H^G, and we have $t_\sigma \geqslant p$, since $t_\sigma \mid |H|$ by Proposition 3.2. Hence, by (3.1),

$$\mathrm{tp}(G) \leqslant P_G(H) = \prod_{\sigma=1}^{s} \frac{t_\sigma!}{t_\sigma^{t_\sigma}} \leqslant \frac{p!}{p^p}.$$

The proof is complete. □

We next give the common transversal probabilities of various groups, information which will be needed later on.

Lemma 4.2. *For an odd prime p and any integer $n \geqslant 1$, let*

$$C_p \rtimes C_{2^n} = \langle a, b \mid a^p = b^{2^n} = 1,\ a^b = a^{-1} \rangle, \tag{4.1}$$

and let D_n be the dihedral group of order $2n$. Then

(i) $\mathrm{tp}(C_p \rtimes C_{2^n}) = (1/2)^{\frac{p-1}{2}}$,

(ii) $\mathrm{tp}(D_n) = (1/2)^{\frac{n-1}{2}}$ *if n is odd, and* $\mathrm{tp}(D_n) = (1/2)^{\frac{n-2}{2}}$ *if n is even,*

(iii) $\mathrm{tp}(Q_{16}) = \mathrm{tp}(C_4 \rtimes C_4) = 1/2$,

(iv) $\mathrm{tp}(A_4) = 2/9$,

(v) $\mathrm{tp}(A_5) = (1/2)^{14}$,

(vi) $\mathrm{tp}(\mathrm{PSL}_3(2)) = (1/2)^{40}$,

(vii) $\mathrm{tp}(C_2^3 \rtimes C_7) = (1/2)^{12}$,

(viii) $\mathrm{tp}(C_3^2 \rtimes C_4) = (1/2)^8$,

(ix) $\mathrm{tp}(\mathrm{SL}_2(3)) = (2/9)^2$.

Proof. (i) Let $G_p = C_p \rtimes C_{2^n}$ be as in (4.1). Then the only subgroups of G_p that are not normal are the conjugates of $\langle b \rangle \cong C_{2^n}$. To see this, observe that a centralises b^2. This is because $a^b = a^{-1}$ implies that $aba = b$, and thus $a^k b a^k = b$ for every $k \geqslant 1$. Hence,

$$ab^2a^{-1} = (aba^{-1})^2 = (ba^{p-2})^2 = ba^{p-2}ba^{p-2} = b^2.$$

Thus, $\langle b^2 \rangle$, which is the unique maximal subgroup of $\langle b \rangle$, is normal in G. Since $\langle b^2 \rangle \cong C_{2^{n-1}}$ is cyclic, all of its subgroups (that is, all proper subgroups of $\langle b \rangle$) are also normal in G_p. Now let $H \leqslant G_p$ be an arbitrary subgroup, and let S be a 2-Sylow subgroup of H. If $S \leqslant \langle b^2 \rangle$, then by the above, S is normal in G_p, and we have $H = S$ or $H = S\langle a \rangle$, so $H \trianglelefteq G_p$ in both cases. There remains the case that $S = \langle b \rangle^x$ for some $x \in G_p$. Then either $H = S = \langle b \rangle^x$ is a conjugate of $\langle b \rangle$, or we have $H = G_p$, which is again a normal subgroup of G_p. Our claim follows. Hence, by Corollary 3.16, we have $\mathrm{tp}(G_p) = P_{G_p}(\langle b \rangle)$.

Now note that $N_{G_p}(\langle b \rangle) = \langle b \rangle$, while for any $x \in G_p$, we get $\langle b \rangle^x \cap \langle b \rangle = \langle b^2 \rangle$. Hence, $|\langle b \rangle : \langle b \rangle^x \cap \langle b \rangle| = 2$ and the $t-$vector of $\langle b \rangle$ is

$$(\underbrace{2, 2, \ldots, 2}_{(p-1)/2}, 1).$$

Hence, $P_{G_p}(\langle b \rangle) = (1/2)^{\frac{p-1}{2}}$ and the first part of the lemma follows.

(ii) We let

$$G = \langle a, b \mid a^2 = b^n = 1,\, b^a = b^{-1} \rangle \cong D_n$$

and work by induction on n. The claim is true for $n = 3$ and $n = 4$ according to a GAP computation [24]. Suppose that the claim has been established for all positive integers j such that $3 \leqslant j < n$, and let K be a non-normal subgroup of $G = D_n$. If $|K| > 2$, then K is dihedral and thus $K \cong D_k$ for some non-trivial proper divisor k of n. Moreover, let

$$N := \mathrm{core}_G(K) = K \cap C,$$

where $C = \langle b \rangle$ is the normal cyclic subgroup of index 2 in D_n generated by a rotation by $2\pi/n$ degrees. Since $G/N = \langle b^{n/k}, a \rangle \cong D_{n/k}$, Corollary 3.17 plus the induction hypothesis yield

$$P_G(K) = P_{G/N}(K/N) \geqslant \mathrm{tp}(D_{n/k}) = \left(\frac{1}{2}\right)^{\frac{n/k-i}{2}}, \quad i = \begin{cases} 1, & n/k \equiv 1\ (2), \\ 2, & n/k \equiv 0\ (2) \end{cases}.$$

On the other hand, $\mathrm{Aut}(D_n)$ acts transitively on the (non-normal) involutions of D_n. To see this, recall that D_n is generated by a rotation r and a reflection s subject to the relation $r^s = r^{-1}$. Now consider the map $\phi_i \colon D_n \to D_n$ fixing every rotation and mapping sr^j to sr^{i+j}, where the exponents work modulo n. Then ϕ_i is a homomorphism (the reader is invited to check this detail), and since it is injective, it is an automorphism of D_n. Finally, notice that if sr^k, sr^ℓ are reflections, then $\phi_\ell \circ \phi_k^{-1}$ sends sr^k to sr^ℓ.

Thus, $P_G(H)$ has the same value for all non-normal subgroups H of G of order 2 by Corollary 3.16. Let H be one such subgroup. Then

$$P_G(H) = \begin{cases} (\frac{1}{2})^{\frac{n-1}{2}} & \text{if } n \text{ is odd}, \\ (\frac{1}{2})^{\frac{n-2}{2}} & \text{if } n \text{ is even}, \end{cases}$$

by Corollary 3.7, where the distinction between cases is explained by the fact that H is self-normalising if n is odd (and thus $m = 1$ in that case), while

$$m = (N_G(H) : H) = 2 \quad \text{if } n \text{ is even}.$$

Since $\mathrm{tp}(D_{n/k}) > P_G(H)$ in either case, $\mathrm{tp}(G)$ is attained for a non-normal cyclic subgroup of order 2, and the induction is complete.

For parts (iii)–(ix) of Lemma 4.2, we use a GAP routine to compute $\mathrm{tp}(G)$.[1] □

[1] The web address https://fourier.math.uoc.gr/~marial/tp-Code contains our GAP code for computing tp(−).

Our next result connects the function $\mathrm{tp}(-)$ to solubility of the corresponding finite group.

Proposition 4.3. *Let G be a finite group such that $\mathrm{tp}(G) > (1/2)^{40}$. Then G is soluble, or has a section isomorphic to A_5. Moreover, this result is best possible.*

Proof. We argue by induction on the order of G. Our claim is obviously true if G is the trivial group so that our induction starts. Assume now that $|G| \geqslant 2$, that $\mathrm{tp}(G) > (1/2)^{40}$, that G is non-abelian, and that our claim holds for groups of smaller order. Suppose first that G is a non-abelian simple group, and let H be a subgroup of order 2 in G (existence of H is guaranteed by the celebrated Feit–Thompson Odd Order Theorem [8]). Then, by Corollary 3.7 and our hypothesis,

$$P_G(H) = \left(\frac{1}{2}\right)^{\frac{n-m}{2}} \geqslant \mathrm{tp}(G) > \left(\frac{1}{2}\right)^{40},$$

where $n = (G : H)$ and $m = (N_G(H) : H)$. Thus, $\frac{n-m}{2} < 40$, or equivalently $n - m < 80$. It follows that

$$|G| - |N_G(H)| < 160. \tag{4.2}$$

Since G is non-abelian simple, no proper subgroup can have index less than 5, so $|N_G(H)| \leqslant \frac{|G|}{5}$, and thus

$$|G| - |N_G(H)| \geqslant \frac{4|G|}{5}. \tag{4.3}$$

Combining (4.3) and (4.2) yields $\frac{4|G|}{5} < 160$, or equivalently $|G| < 200$. However, the only non-abelian simple groups of order less than 200 are A_5 (of order 60) and $\mathrm{PSL}_3(2)$ (of order 168). Now A_5 satisfies $\mathrm{tp}(A_5) = (1/2)^{14} > (1/2)^{40}$ by part (v) of Lemma 4.2, but trivially has a section isomorphic to A_5, thus is acceptable, while $\mathrm{tp}(\mathrm{PSL}_3(2)) = (1/2)^{40}$, so that $G \cong \mathrm{PSL}_3(2)$ is ruled out by our hypothesis. Thus, $G \cong A_5$ for G non-abelian simple, and the claim follows in that case.

Consequently, we may assume further that G is not simple, thus G has a proper non-trivial normal subgroup N. By parts (i) and (ii) of Proposition 4.1, we have $\mathrm{tp}(N) \geqslant \mathrm{tp}(G) > 2^{-40}$ and $\mathrm{tp}(G/N) \geqslant \mathrm{tp}(G) > 2^{-40}$, thus, by the induction hypothesis applied to N and G/N, each of N and G/N is either soluble or has a section isomorphic to A_5. If both are soluble, so is G. If N has a section isomorphic to A_5, then so does G, while if G/N has a section isomorphic to A_5, this section lifts to give a corresponding section isomorphic to A_5 in G. This completes the induction. Finally, since $G = \mathrm{PSL}_3(2)$ is not soluble (in fact, G is non-abelian simple), $5 \nmid |G|$ (so that G cannot contain a section isomorphic to A_5), and $\mathrm{tp}(G) = (1/2)^{40}$ by part (vi) of Lemma 4.2, our result is indeed best possible. □

We now turn our attention to the connection of $\mathrm{tp}(-)$ and supersolubility. The proof of Theorem 4.4 below is considerably harder than that of Proposition 4.3, which is mainly due to the fact that the class of supersoluble groups is not closed under extensions (otherwise every metabelian group would be supersoluble, but A_4 is a counterexample).

Theorem 4.4. *Let G be a finite group with $\mathrm{tp}(G) > (1/2)^8$. Then either G is supersoluble, or G has a section isomorphic to A_4. Moreover, the bound $(1/2)^8$ is sharp.*

Proof. We argue by induction on the order of G. If G is trivial, then the theorem clearly holds so that the induction starts. Suppose that $|G| \geqslant 2$, that G is non-abelian, that $\mathrm{tp}(G) > 2^{-8}$, and that the result holds for groups of smaller order. Since $\mathrm{tp}(G) > (1/2)^8 > (1/2)^{40}$, our group G is soluble by Proposition 4.3. Let A be a minimal normal subgroup of G. Then A is an elementary abelian group of order p^r for some prime p and some positive integer r (see, for instance, [15, Chapter I, Satz 9.13]). Since, by part (ii) of Proposition 4.1 plus our hypothesis concerning G, we have

$$(1/2)^8 < \mathrm{tp}(G) \leqslant \mathrm{tp}(G/A),$$

the inductive hypothesis applies to the quotient G/A. Hence, either G/A is supersoluble or G/A has a section isomorphic to A_4. In the latter case, the section in question lifts isomorphically to a section in G; thus, we may suppose that G/A is supersoluble.

If A is cyclic, then G is also supersoluble (see, e.g., [22, Statement 7.2.14]). Thus, we may suppose further that A is not cyclic, that is, $r > 1$ and $A \not\leqslant Z(G)$. If $A \leqslant \Phi(G)$, then G/A supersoluble would imply that G itself is supersoluble owing to the fact that supersoluble groups comprise a saturated formation (see, for instance, [15, Chapter VI, Satz 8.6 (a)]). Consequently, we may also assume that $A \not\leqslant \Phi(G)$ so that there exists a maximal subgroup M of G with $A \not\leqslant M$; in particular, $AM = G$. As A is normal in G, we have

$$(A \cap M)^g = A^g \cap M^g = A \cap M^g = A \cap M \quad (g \in G),$$

where $M = M^g$ comes from the fact that $MM^g \neq G$ by a theorem of Ore's (see [15, Chapter II, Satz 3.9]). Thus, $A \cap M \trianglelefteq G$, and since $A \cap M < A$, minimality of A forces $A \cap M = 1$, so that G splits over A. Furthermore, $G/A \cong M$, and thus M is supersoluble. Let T be a minimal normal subgroup of M. Then T is a cyclic group of order q for some prime q, while $N_G(T) \geqslant M$. Since M is maximal in G, either $N_G(T) = M$ or $T \trianglelefteq G$. In the second case, invoking Proposition 4.1 (ii), our assumption that $\mathrm{tp}(G) > (1/2)^8$, and the induction hypothesis, we deduce that

either G/T (and thus G) is supersoluble, or G/T, and thus G, has a section isomorphic to A_4. We may therefore assume that $N_G(T) = M$. If $(M : T) = t$, then Corollary 3.7 implies that

$$\left(\frac{1}{2}\right)^8 < \mathrm{tp}(G) \leqslant P_G(T) = \left(\frac{q!}{q^q}\right)^{\frac{t(p^r-1)}{q}}, \tag{4.4}$$

where we note that

$$(G : M) = (A \rtimes M : M) = |A| = p^r$$

and that the exponent $t(p^r - 1)/q$ is integral (see Corollary 3.7). Suppose first that $q \geqslant 5$. Then $q!/q^q \leqslant 5!/5^5 = 24/625$, and thus

$$\left(\frac{q!}{q^q}\right)^2 < \left(\frac{1}{2}\right)^8 < \mathrm{tp}(G) \leqslant P_G(T) = \left(\frac{q!}{q^q}\right)^{\frac{t(p^r-1)}{q}}.$$

The above inequality forces the exponent of the right-hand-side to be equal to 1 so that $q = t(p^r - 1)$. As $r > 1$, we conclude that $t = 1$ and so $p = 2$ and q is a Mersenne prime (thus r is prime as well). But the only Mersenne prime q such that $q!/q^q > (1/2)^8$ is $q = 7$, which implies that $|G| = 56$. Using GAP, we see that the only non-supersoluble group of order 56 is the Frobenius group $C_2^3 \rtimes C_7$, whose transversal probability is $1/4096 < 2^{-8}$ by part (vii) of Lemma 4.2.

Now suppose that $q = 3$. Then

$$\left(\frac{3!}{3^3}\right)^4 < \left(\frac{1}{2}\right)^8 < \mathrm{tp}(G) \leqslant P_G(T) = \left(\frac{3!}{3^3}\right)^{\frac{t(p^r-1)}{3}}.$$

So $t(p^r - 1)/3 \leqslant 3$. As $r > 1$ and $(p^r - 1) \geqslant 3$, we conclude that one of the following occurs:

$$\begin{cases} t = 1 \text{ and } p^r = 4, 8, 9, \\ t = 2 \text{ and } p^r = 4, \\ t = 3 \text{ and } p^r = 4. \end{cases}$$

So, in all cases, $t \cdot p^r \leqslant 12$, and thus $|G| = 3 \cdot t \cdot p^r \leqslant 36$. Using GAP, we see that the non-supersoluble groups of order at most 36 satisfying $\mathrm{tp} > (1/2)^8$ are $A_4, \mathrm{SL}_2(3), S_4, C_2 \times A_4, C_3.A_4$, and $C_3 \times A_4$, all of which have a section isomorphic to A_4 (actually, all but S_4 have a quotient isomorphic to A_4).

We should mention here that, among the non-supersoluble groups of order 36, there is the group $G = C_3^2 \rtimes C_4$, whose transversal probability is exactly $(1/2)^8$ and G contains no section isomorphic to A_4; see part (viii) of Lemma 4.2. This implies that our bound, when proved, will be sharp.

We are left with the case where $q = 2$. Clearly, for (4.4) to hold, we must have $t(p^r - 1)/2 \leqslant 7$. We still have $r > 1$ and $p^r - 1 \geqslant 3$; thus, we get one of the following cases:

$$\begin{cases} t = 1 \text{ and } p^r = 4, 8, 9, \\ t = 2 \text{ and } p^r = 4, 8, \\ t = 3 \text{ and } p^r = 4, \\ t = 4 \text{ and } p^r = 4. \end{cases}$$

In all cases, we have $t \cdot p^r \leqslant 16$; thus, $|G| = 2 \cdot t \cdot p^r \leqslant 32$. As we have already checked, there is no counterexample to our claim among the groups of order at most 36, so the proof is complete. □

The last theorem of this subsection relates $\mathrm{tp}(-)$ and nilpotency. In order to express ourselves concisely and avoid repetition, we adopt a local convention. We will say that the group G has a *bad section* if it has a section isomorphic to one of the groups in the following set: $\{A_4, D_3, D_5, D_7\}$.

Theorem 4.5. *Let G be a finite group with $\mathrm{tp}(G) > 4/81 = (2/9)^2$. Then either G is nilpotent, or G has a bad section. Furthermore, the bound is sharp.*

Proof. We argue by induction on the order of G. If G is trivial, then the theorem clearly holds; thus, the induction starts. Suppose that $|G| \geqslant 2$, that $\mathrm{tp}(G) > 4/81$, and that the result holds for groups of smaller order. Since $\mathrm{tp}(G) > 4/81 > (1/2)^8$, Theorem 4.4 implies that either G is supersoluble, or has a section isomorphic to A_4. Thus, we may assume that our group is supersoluble.

We now argue that $\Phi(G) = 1$ and that G has a unique minimal normal subgroup, say A, which must moreover be cyclic of prime order, since G is supersoluble.

- To justify the first assertion, suppose instead that $\Phi(G) > 1$. Then

  $$4/81 < \mathrm{tp}(G) \leqslant \mathrm{tp}(G/\Phi(G)),$$

  and thus the inductive hypothesis applied to $G/\Phi(G)$ gives that $G/\Phi(G)$ is nilpotent, or that $G/\Phi(G)$ has a bad section. If $G/\Phi(G)$ is nilpotent, then G is nilpotent, and we are done. If, on the other hand, the group $G/\Phi(G)$ has a bad section, then the section in question lifts isomorphically to a section in G; thus, G itself has a bad section, and we are done in that case as well.
- To justify the second assertion, suppose that there exist minimal normal subgroups $A \neq B$. Then G/A may be assumed to be nilpotent, and the same holds for G/B. But nilpotent groups comprise a formation; thus, $G/(A \cap B)$ is nilpotent; hence, G itself is nilpotent (since $A \cap B = 1$), completing the proof.

So let A be the unique minimal normal subgroup of G. Since $\Phi(G) = 1$, there exists a maximal subgroup M not containing A, and thus $A \cap M = 1$. It follows that M complements A in G; hence, $G = A \rtimes M$. Now observe that M is necessarily core-free owing to the fact that A is the unique minimal normal subgroup of G. Then $C_G(A)$ is normal in G and contains A. By Dedekind's lemma, we have $C_G(A) = AK$, where $K := C_M(A)$. Thus, K is normal in M and is centralised by A; hence, $K \trianglelefteq G$. It follows that K is trivial and A is self-centralising. By the N/C-theorem, $M \cong G/A$ embeds isomorphically as a subgroup of the automorphism group of A, which is cyclic of order $p-1$. Moreover, every non-trivial element of A is a generator of A; thus, for all $a \in A$ with $a \neq 1$, we have

$$C_G(a) = C_G(\langle a \rangle) = C_G(A) = A \leqslant A.$$

Thus (see [16, Theorem 6.4]), G is isomorphic to a Frobenius group, with A the Frobenius kernel and M a Frobenius complement being cyclic of order d, where $d \mid p-1$. By Corollary 3.6, we have

$$P_G(M) = \left(\frac{d!}{d^d}\right)^{\frac{p-1}{d}} \geqslant \mathrm{tp}(G) > \frac{4}{81}.$$

If $d = p-1$, then the fact that $d!/d^d > 4/81$ forces $d \leqslant 4$, and thus $d \in \{1, 2, 4\}$. Then G is isomorphic to C_2 or to D_3, or to the Frobenius group of order 20, respectively, so G is nilpotent or else G has a bad section (in the last case because it has a subgroup isomorphic to D_5). We may therefore assume that $d < p-1$; hence, $(p-1)/d \geqslant 2$. Thus,

$$\left(\frac{d!}{d^d}\right)^2 \geqslant \left(\frac{d!}{d^d}\right)^{\frac{p-1}{d}} > \left(\frac{2}{9}\right)^2,$$

forcing $d!/d^d > 2/9$. That is only possible if $d = 2$, whence G is itself a dihedral group. But the inequality

$$\left(\frac{1}{2}\right)^{\frac{p-1}{2}} > \left(\frac{2}{9}\right)^2$$

only holds for $p \leqslant 7$, in which case G has a bad section.

Finally, the semidirect product $C_7 \rtimes C_3$ has common transversal probability equal to $4/81$ and is neither nilpotent nor does it contain a bad section. Thus, our result is indeed sharp, and the proof is complete. □

4.2 Structural characterisation of certain tp-values

We saw in the previous section how the knowledge of the range of values of the invariant $\mathrm{tp}(G)$ provides structural information for the group G. But what can be said about specific values of $\mathrm{tp}(G)$? For example, we have seen that if G

is not a Dedekind group, then there exists a non-normal subgroup H, and thus we have $\mathrm{tp}(G) \leqslant P_G(H) \leqslant 1/2$, so it is natural to wonder whether we might be able to characterise all groups with $\mathrm{tp}(G) = 1/2$. We will actually show that if $\mathrm{tp}(G) = p!/p^p$ for a prime p, then $p = 2$ and the group G is one of a very specific list of groups (cf. Corollary 4.11 below). We will also show that, like the value $p!/p^p$ for p an odd prime which cannot be attained, also tp can never equal $p!/p^p \cdot q!/q^q$ for distinct primes $p < q$ (see Theorem 4.8 below). On the other hand, if $p = q = 2$, we may completely characterise groups with transversal probability $\mathrm{tp}(G) = 1/4$ (cf. Theorem 4.13).

Our first result is purely arithmetic, and is of independent interest.

Theorem 4.6. *Assume that*

$$\prod_{i=1}^{n} \frac{t_i!}{t_i^{t_i}} = \prod_{i=1}^{k} \frac{p_i!}{p_i^{p_i}}, \tag{4.5}$$

where the numbers $t_i > 1$ are positive integers and the p_i are distinct primes. Then $k = n$ and, after appropriate rearrangement, $t_i = p_i$ for all $i = 1, \ldots, n$.

Proof. We argue by induction on k. Assume $k = 1$, that is,

$$\prod_{i=1}^{n} \frac{t_i!}{t_i^{t_i}} = \frac{p!}{p^p}.$$

Then there exists a t_σ such that $p \mid t_\sigma$ and, without loss of generality, we may assume that $p \mid t_1$. Now, the sequence $a_n = n!/n^n$ is monotonically decreasing, as

$$\frac{a_{n+1}}{a_n} = \Big(\frac{n}{n+1}\Big)^n = \Big(\frac{1}{1+1/n}\Big)^n < 1,$$

while clearly $a_n < 1$ for all $n > 1$. Hence

$$\frac{p!}{p^p} = \prod_{i=1}^{n} \frac{t_i!}{t_i^{t_i}} \leqslant \frac{t_1!}{t_1^{t_1}} \leqslant \frac{p!}{p^p}.$$

Therefore, $t_1 = p$ and $n = 1$.

Assume now that the inductive hypothesis holds for all values less than k; we shall prove it for k. Thus, we have

$$\prod_{i=1}^{n} \frac{t_i!}{t_i^{t_i}} = \prod_{i=1}^{k} \frac{p_i!}{p_i^{p_i}}$$

for distinct primes $\{p_i\}_{i=1}^{k}$, and we assume that $\{t_i\}$ and $\{p_i\}$ are written in a decreasing order. So $p := p_1$ is the largest prime involved in the right-hand side, and

we have $t_1 \geqslant t_2 \geqslant \cdots \geqslant t_n$. Clearly, $p > 2$, and equation (4.5) implies that

$$p^{p-1} \prod_{i=2}^{k} p_i^{p_i} \prod_{i=1}^{n} t_i! = (p-1)! \prod_{i=2}^{k} p_i! \prod_{i=1}^{n} t_i^{t_i}.$$

Hence, p^{p-1} divides $\prod_{i=1}^{n} t_i^{t_i}$, and thus p divides at least one of the t_i's. If $p = t_\sigma$ for some $\sigma = 1, \ldots, n$, then we are done by the inductive hypothesis. To see this, divide both sides of (4.5) by $p!/p^p$ to get

$$\prod_{i=1, i\neq\sigma}^{n} \frac{t_i!}{t_i^{t_i}} = \prod_{i=2}^{k} \frac{p_i!}{p_i^{p_i}},$$

where the inductive hypothesis now applies.

So we may assume that $t_i \neq p$ for all $i = 1, \ldots, n$, while the fact that p divides some t_i implies that $t_1 \geqslant 2p$.

For any $x = m/n \in \mathbb{Q}$, let $v_p(x)$ denote the p-adic valuation of $x = m/n$, that is, if $x \in \mathbb{Z}$, then $v_p(x)$ is the exponent of the largest power of p that divides x, and $v_p(x) = 0$ if $p \nmid x$, while for $x = m/n$, we write

$$v_p(x) = v_p(m/n) = v_p(m) - v_p(n).$$

Then observe that, for any integer t, we have that

$$v_p(t!/t^t) < 0 \quad \text{if and only if} \quad p \mid t.$$

Now let q be the biggest prime smaller than t_1. Then $q > p$ because $t_1 \geqslant 2p$ and, according to Bertrand's postulate, there exists a prime in the interval $(p, 2p)$, cf., for instance, [14, Theorem 418]. Hence,

$$p < q \leqslant t_1.$$

Assume now that $t_1 = q$ and observe that $v_q(t_i!/t_i^{t_i}) = 1 - q$ for all i with $t_i = q$ (there is at least one such, namely t_1), while for the remaining, $v_q(t_i!/t_i^{t_i}) = 0$, since t_1 is the biggest of the t_i's. Hence, equation (4.5) implies that

$$\sum_{t_i=q} (1-q) = \sum_{i=1}^{n} v_q\Big(\frac{t_i!}{t_i^{t_i}}\Big) = \sum_{i=1}^{k} v_q\Big(\frac{p_i!}{p_i^{p_i}}\Big) = 0,$$

which is clearly absurd. Thus, $q < t_1$. In addition, applying Bertrand's postulate again for the prime q, we conclude that $t_1 < 2q$ (or else a bigger prime than q exists in $(q, 2q)$ and this prime would be smaller than t_1 contradicting the choice of q). Hence,

$$p < q < t_1 < 2q.$$

Therefore, $q \nmid t_1$. So $v_q(t_1!/t_1^{t_1}) = 1$, while

$$v_q\left(\prod_{i=1}^{k} \frac{p_i!}{p_i^{p_i}}\right) = 0$$

as $q > p$ and p is the largest of the p_i's.

Thus, in view of (4.5), we get

$$1 + \sum_{i=2}^{n} v_q\left(\frac{t_i!}{t_i^{t_i}}\right) = v_q\left(\prod_{i=1}^{k} \frac{p_i!}{p_i^{p_i}}\right) = 0.$$

We conclude that

$$\sum_{i=2}^{n} v_q\left(\frac{t_i!}{t_i^{t_i}}\right) < 0.$$

Hence, there exists t_s, for some $s = 2, \ldots, n$ so that $q \mid t_s$. But $t_i \leqslant t_1 < 2q$ for every $i = 1, \ldots, n$. Hence, there exists some $s \in \{2, \ldots, n\}$ with $t_s = q$, and thus $v_q(t_s!/t_s^{t_s}) = 1 - q$. We conclude that

$$2 - q + \sum_{i=2, i\neq s}^{n} v_q\left(\frac{t_i!}{t_i^{t_i}}\right) = 0.$$

Hence, there exist at least $q - 2$ among the t_i's that are greater than q. These along with t_1 and t_s provide at least q elements among the t_i's that are greater than or equal to q. Hence, $t_i \geqslant q$ for all $i = 1, \ldots, q$ (as these are greater than or equal to the q-previously picked t_i's) and therefore

$$\sum_{i=1}^{q} t_i \geqslant q^2 > p^2 > \frac{p(p+1)}{2},$$

while

$$\sum_{i=1}^{k} p_i \leqslant \sum_{k=1}^{p_1} k = \frac{p(p+1)}{2}.$$

We conclude that the vector $(t_1, t_2, \ldots, t_q, \ldots, t_n)$ weakly majorises the vector $(p_1, p_2, \ldots, p_k)$, while all the hypothesis of Proposition A.6 (see appendix) are satisfied. Hence,

$$\prod_{i=1}^{n} \frac{t_i!}{t_i^{t_i}} < \prod_{i=1}^{k} \frac{p_i!}{p_i^{p_i}},$$

contradicting the hypothesis of the theorem. This final contradiction implies that there does exist t_i with $t_i = p$ and so, as we have seen, the inductive hypothesis completes the proof of the theorem. □

Combining Theorem 4.6 with Proposition 3.2, we get the following.

Corollary 4.7. *Assume that*

$$P_G(H) = \frac{p!}{p^p} \cdot \frac{q!}{q^q}$$

with distinct primes $p < q$, where G is a finite group with subgroup H. Then the t-vector of H in G is given by $(q, p, 1, \ldots, 1)$; in particular, $pq \mid |H|$.

With the help of Corollary 4.7, we can now prove a result excluding certain rational numbers from the range of the function $\mathrm{tp}(-)$.

Theorem 4.8. *There exists no finite group G such that $\mathrm{tp}(G) = p!/p^p \cdot q!/q^q$ for primes $p < q$.*

Proof. Let G be a counterexample of smallest possible order, and let H be a proper subgroup of G for which $\mathrm{tp}(G)$ is realised, that is,

$$P_G(H) = \mathrm{tp}(G) = \frac{p!}{p^p} \cdot \frac{q!}{q^q}$$

for distinct primes $p < q$. We argue that H is core-free. Suppose not, and let $N := \mathrm{core}_G(H)$. Then $N > 1$ and $P_{G/N}(H/N) = P_G(H)$ by Corollary 3.17. Moreover, by part (ii) of Proposition 4.1, we have

$$P_{G/N}(H/N) \geqslant \mathrm{tp}(G/N) \geqslant \mathrm{tp}(G) = P_G(H).$$

Therefore, $\mathrm{tp}(G/N) = \mathrm{tp}(G)$, contradicting the minimality of G. Thus, H is core-free, as claimed.

The prime q divides $|H|$ according to Corollary 4.7. Let $K < H$ be of order q. Then K is not normal in G, as H is core-free. Also

$$P_G(K) = \left(\frac{q!}{q^q}\right)^{\frac{n-m}{q}},$$

in view of Corollary 3.7, where $n = (G : K)$, $m = (N_G(K) : K)$, $m \mid n$, and $m \neq n$. In addition, we have

$$\left(\frac{q!}{q^q}\right)^{\frac{n-m}{q}} = P_G(K) \geqslant \mathrm{tp}(G) = \frac{p!}{p^p} \cdot \frac{q!}{q^q}.$$

But $p < q$, and thus $p!/p^p > q!/q^q$. So the above inequality holds if and only if $\frac{n-m}{q} = 1$, which in turn, as $m \mid n$, implies that $m \mid q$. Hence, either $m = 1$ and $n = q + 1$, or $m = q$ and $n = 2q$.

In the first case, we get $K = N_G(K)$, and thus we have $K = N_H(K)$, while $|G| = q(q+1)$. Furthermore, by Proposition 3.14 and Corollary 3.7, we have

$$\frac{q!}{q^q} = P_G(K) \leqslant P_H(K) = \left(\frac{q!}{q^q}\right)^{\frac{(H:K)-1}{q}}.$$

Hence, $0 \leqslant \frac{(H:K)-1}{q} \leqslant 1$. As $H \neq K$, we necessarily have $(H : K) - 1 = q$, and thus $(H : K) = q + 1 = (G : K)$ or, equivalently, $G = H$, contradicting the fact that we had chosen H as a proper subgroup of G. So the first case, where $m = 1$, does not occur.

Consequently, we must have $m = q$, $n = 2q$, $|G| = 2q^2$. This forces $p = 2$, the only other prime involved in the order of G. As $H < G$, while 2 and q both divide H by Corollary 4.7, we get $|H| = 2q$; thus, K is a normal subgroup of H. So $H \leqslant N_G(K)$, and thus

$$2 = (H : K) \mid (N_G(K) : K) = q.$$

This final contradiction now implies the theorem. □

We can now show that if, for a specific subgroup $H \leqslant G$, we know that

$$P_G(H) = \frac{p!}{p^p}$$

for some prime p, then H has a very restricted position inside G.

Lemma 4.9. *Let G be a finite group and $H \leqslant G$ a subgroup. If $P_G(H) = p!/p^p$ for some prime p, then $p \mid |H|$ and one of the following occurs:*

(i) $H = N_G(H)$ *and* $(G : H) = p + 1$, *or*

(ii) $(N_G(H) : H) = p$, *while* $(G : H) = 2p$.

In particular, $P_G(H) = 1/2$ implies that H is a subgroup of even order, whose index in G is either 3 *or* 4.

Proof. Suppose that $P_G(H) = p!/p^p = \prod_\sigma t_\sigma!/t_\sigma^{t_\sigma}$ for some prime p, and let $m = (N_G(H) : H)$. Combining Proposition 3.2 with Theorem 4.6, we see that Γ_H^G has precisely one non-trivial component $\Delta_{\sigma_1} \cong K_{t_{\sigma_1}, t_{\sigma_1}}$, and that $t_{\sigma_1} = p$. Hence, $p \mid |H|$, and we have

$$n = (G : H) = m + p.$$

Since $m \mid n$, we get $m \mid p$. Thus, either $m = 1$ (i.e., $N_G(H) = H$) and $n = p + 1$, or $m = p$ and $n = 2p$, giving $(N_G(H) : H) = p$ in the second case. □

Our next result takes a major step towards classifying groups G where

$$\mathrm{tp}(G) = \frac{p!}{p^p}$$

for some prime p.

Theorem 4.10. *Let G be a finite group. Assume that, for every non-normal subgroup $H \leqslant G$, we have $P_G(H) = p!/p^p$ with some fixed prime p. Then one of the following holds.*

(i) *G is a Dedekind group. In this case, all subgroups are normal and* $\mathrm{tp}(G) = 1$.

(ii) $G \cong C_3 \rtimes C_{2^n}$ *for some integer* $n \geqslant 1$.

(iii) $G \cong D_4$.

(iv) $G \cong Q_{16}$.

(v) $G \cong C_4 \rtimes C_4$.

In cases (ii)–(v)*, we have $p = 2$ and* $\mathrm{tp}(G) = 1/2$.

Proof. Clearly, $\mathrm{tp}(G) = 1$ if and only if every subgroup of G is normal or, equivalently, if and only if G is a Dedekind group. Thus, discarding case (i), we assume from now on that $\mathrm{tp}(G) < 1$.

Fix some non-normal subgroup H of G. Then, by hypothesis,

$$P_G(H) = \frac{p!}{p^p},$$

and thus, by Lemma 4.9, we have $(G : H) = p + 1$ or $(G : H) = 2p$, as well as $p \mid |H|$.

Claim. *Every proper subgroup of H is normal in G and $H \cong C_{p^k}$, a cyclic group of order p^k for some positive integer k.*

Proof of claim. If $(G : H) = p + 1$ and $K < H$, then clearly $(G : K) \neq p + 1$. Also, $(G : K) \neq 2p$, or else $2p = r(p + 1)$ for some integer $r \geqslant 2$. This implies that $p \mid r$, while $2 = r + r/p > 2$, a contradiction. If $(G : H) = 2p$ and $K < H$, then $(G : K) \geqslant 4p$; in particular, $(G : K) \neq p + 1, 2p$. Hence, in both cases, K is normal, and the first part of the claim follows.

If M_1, M_2 are distinct maximal subgroups of H, then they are normal in G, and their product is $M_1 M_2 = H$. This would imply that H is normal in G, contrary to our assumption. Hence, H has a unique maximal subgroup; thus, H is cyclic of prime power order. Moreover, as $p \mid |H|$, we conclude that $H \cong C_{p^k}$ for some $k \geqslant 1$, and our claim follows. □

For the rest of the proof, we shall need to distinguish two cases.

Case 1: *p is odd.* As we have seen, either $(G : H) = p + 1$ and $H = N_G(H)$, or $(G : H) = 2p$ and $|N_G(H)| = p^{k+1}$. Hence, we either have $|G| = p^k(p + 1)$ or $|G| = 2p^{k+1}$. Therefore, in either case, $2 \mid |G|$ so that there exists some $x \in G$ of order 2. Let $T = \langle x \rangle$. Clearly, T is not a subgroup of $N_G(H)$ (in either case). This implies that T cannot be normal in G. Indeed, if $T \trianglelefteq G$, then $x \in Z(G)$, and so T would be a subgroup of $N_G(H)$. It follows that $p \mid |T| = 2$, forcing $p = 2$, which contradicts our case assumption. Consequently, p has to be even.

Case 2: $p = 2$. Here, there are two subcases:

(a) $N_G(H) = H$ and $(G : H) = 3$, or

(b) (b) $(N_G(H) : H) = 2 = (G : N_G(H))$.

Subcase (a). By our claim proved above, $H \cong C_{2^n}$ for some positive integer n. Since $(G : H) = 3$ so that $|G| = 3 \cdot 2^n$, it follows that if Q is a Sylow 3-subgroup of G, then $Q \cong C_3$. Moreover, Q is normal in G, since otherwise $P_G(Q) = 1/2$, so that Q would have to have even order by Lemma 4.9. By a well-known theorem of Zassenhaus (see, for instance, [15, Hauptsatz I.18.1]), we have

$$G \cong \langle a, b \mid a^3 = b^{2^n} = 1,\ a^b = a^{-1} \rangle \cong C_3 \rtimes C_{2^n}, \quad n \geqslant 1,$$

since $\mathrm{tp}(G) < 1$, whence case (ii) of the theorem.

Subcase (b). Now suppose that

$$(G : N_G(H)) = 2 = (N_G(H) : H),$$

where $H \cong C_{2^n}$ for some $n \geqslant 1$, so that G is a 2-group. At this point, we are tasked with determining the collection of all 2-groups G of order $|G| \geqslant 8$, which enjoy the following property:

all subgroups of G are normal in G except some subgroups of index 4. (†)

Note that we are not allowing for any ambiguity in (†); there *must* exist non-normal subgroups of index 4. The only group of order 8 with this property is clearly D_4 and we use GAP to find the groups of order 16: they are Q_{16} and `SmallGroup(16,4)`, which is of type $C_4 \rtimes C_4$.

We now argue that those are the only 2-groups with this property. To prove the assertion, we shall show that if G satisfies (†), then $|G| \leqslant 16$. Indeed, suppose that G is a minimal counterexample to our last claim, and let H be a non-normal subgroup of index 4 in G. Let N be a subgroup of order 2 in H, and observe that N is normal in G, since $|G| \geqslant 32$, and G satisfies (†). By the Correspondence

Theorem, G/N satisfies (†) and is not a counterexample. Thus, $|G/N| \leqslant 16$, and so $|G| \leqslant 32$, which implies that $|G| = 32$. Using GAP again, we check that no group of order 32 satisfies (†), and we have reached the desired contradiction. Hence, our claim holds, and the only 2 groups satisfying (†) are those exhibited above. □

As a consequence of Theorem 4.10, we can now determine, up to isomorphism, all groups G with $\mathrm{tp}(G) = 1/2$.

Corollary 4.11. *Let G be a finite group. We have $\mathrm{tp}(G) = 1/2$ if and only if one of the following occurs:*

(i) $G = C_3 \rtimes C_{2^n}$ *for some integer* $n \geqslant 1$,

(ii) $G = D_4$,

(iii) $G = Q_{16}$,

(iv) $G = C_4 \rtimes C_4$.

Proof. This follows from Theorem 4.10 in conjunction with Corollary 3.10. □

As a further application of Theorem 4.10, we have the following.

Corollary 4.12. *If G is a non-abelian group of odd order, then $\mathrm{tp}(G) \leqslant 4/81$.*

Proof. Let $H \leqslant G$ be a non-normal subgroup. As

$$m = (N_G(H) : H) < (G : H) = n,$$

the graph Γ_H^G has at least one non-trivial component $\Delta_\sigma \cong K_{t_\sigma, t_\sigma}$, and since $1 < t_\sigma \mid |H| \mid |G|$ by Proposition 3.2, and G has odd order, we have $t_\sigma \geqslant 3$. It follows that, for any such group G, either $\mathrm{tp}(G) \leqslant 4/81$, or $P_G(H) = 2/9$ for all subgroups H of G which are not normal in G. In the second case, Theorem 4.10 applies so that G would have to be one of the groups listed in (i)–(v). However, by the well-known classification of Dedekind groups, case (i) does not apply, while cases (ii)–(v) are ruled out as G has odd order by hypothesis. Hence, $\mathrm{tp}(G) \leqslant 4/81$, as claimed. □

Our next result determines all groups G with $\mathrm{tp}(G) = 1/4$.

Theorem 4.13. *Let G be a finite group. We have $\mathrm{tp}(G) = 1/4$ if and only if one of the following occurs.*

(i) $G = \langle a, b \mid a^5 = b^{2^k} = 1, a^b = a^{-1} \rangle$ *for some integer* $k \geqslant 1$.

(ii) *G has a normal cyclic 2-subgroup $M = C_{2^k}$ for some integer $k \geqslant 0$, so that G/M is one of the groups D_6, $M_4(2)$, $C_4 \circ D_4$, $C_2 \times D_4$, or $C_2^2 \rtimes C_4$.*

Proof. The proof is similar to that of Theorem 4.10 so we only sketch some steps. Assume first that $P_G(H) = 1/4$ for some subgroup H of G, where

$$\frac{1}{4} = \prod_{\sigma=1}^{s} \frac{(t_\sigma)!}{t_\sigma^{t_\sigma}}.$$

Then note that we necessarily have two values $t_{\sigma_1} = t_{\sigma_2} = 2$, while the rest are all 1. The reason is that if there exist $t_\sigma \geqslant 3$, then $(t_\sigma)!/t_\sigma^{t_\sigma} \leqslant 2/9$ and thus

$$\prod_{\sigma=1}^{s} \frac{(t_\sigma)!}{t_\sigma^{t_\sigma}} \leqslant \frac{2}{9}.$$

Hence, $2 \mid |H|$ and $4 + m = n$, where m, n are, as usual, the index of H in $N_G(H)$ and in G, respectively. Furthermore, $m \mid n$, and thus $m \mid 4$. Hence, $m = 1, 2$, or 4. In conclusion, if $P_G(H) = 1/4$, then H is group of even order whose indices m, n are either

$$m = 1 \text{ and } n = 5, \quad m = 2 \text{ and } n = 6, \quad \text{or} \quad m = 4 \text{ and } n = 8.$$

Assume now that $\mathrm{tp}(G) = 1/4$. Then, for every non-normal subgroup T of G, we should have $P_G(T) = 1/4$ or $P_G(T) = 1/2$. Hence, the only possible values for the index $(G : T)$ are $3, 4, 5, 6$, and 8. (The first two values occur in the case that $P_G(T) = 1/2$, according to Lemma 4.9.) As $\mathrm{tp}(G) = 1/4$, there exists a non-normal subgroup $H \leqslant G$ with $P_G(H) = 1/4$, and thus $(G : H) \in \{5, 6, 8\}$, while H has even order. Observe that if M is any proper subgroup of H, then $(G : M) \notin \{3, 4, 5, 6, 8\}$. Therefore, any proper subgroup of H is normal in G. We conclude that $H = C_{2^k}$ for some integer $k \geqslant 1$. We now distinguish three cases.

Case 1: $(G : H) = 5$ *and* $H = N_G(H)$. If C is a 5-Sylow subgroup of G, then $P_G(C) = 1$, as $|C| = 5$ is odd. Hence, C is a normal subgroup of G. If M is the maximal subgroup of H, then $C \cdot M = C \times M$. So M is a central subgroup of G. Let $C = \langle a \mid a^5 = 1 \rangle$ and $H = \langle b \mid b^{2^k} = 1 \rangle$. Then b acts as an automorphism of order 2 on C. We conclude that

$$G = \langle a, b \mid a^5 = b^{2^k} = 1,\ a^b = a^{-1} \rangle.$$

According to Lemma 4.2 (i), the above group has transversal probability equal to 1/4 and Case 1 is completed.

Case 2: $(G : H) = 6$ *and* $(N_G(H) : H) = 2$. In this case the, 3-Sylow subgroup C of G is a normal subgroup of G, or else we would have $P_G(C)$ being a power of $2/9$. If $|H| = 2$, then $G = D_6$. So we may assume that $H = C_{2^k}$ with $k > 1$.

Let $M = \langle b^2 \rangle$ be the maximal subgroup of H. Then $1 \neq M \trianglelefteq G$. In addition, $1/4 = P_G(H) = P_{G/M}(H/M)$, while $\mathrm{tp}(G) \leqslant \mathrm{tp}(G/M)$. We conclude that G/M is a group of order 12 whose common transversal probability is $1/4$. Using GAP, we find that $G/M = D_6$.

Case 3: $(G : H) = 8$ *and* $(N_G(H) : H) = 4$. In this case, G is a 2-group. If $|H| = 2$, then G is a group of order 16. Using GAP again, we see that the only groups of order 16 that have transversal probability $1/4$ are $M_4(2)$, $C_4 \circ D_4$, $C_2 \times D_4$, and $C_2^2 \rtimes C_4$. We may assume $|H| > 2$, and we write M for its unique maximal subgroup. Then M is normal in G. As earlier, we get $\mathrm{tp}(G/M) = 1/4$, while $|G/M| = 16$. Hence, G/M is one of the previously mentioned groups.

This completes the proof in Case 3, and the theorem follows. □

Proposition 4.3 might have left the reader (as it did us) with the nagging feeling that a refinement might perhaps be possible, which connects $\mathrm{tp}(-)$ with the derived length of a soluble group. The final result of this section is a first indication that such a result might indeed exist, although we have been unable so far to find it. The reader should also note that the bound obtained in the next result is the same one achieved in Theorem 4.5.

Proposition 4.14. *If G is a non-abelian group with $\mathrm{tp}(G) > 4/81$, then G has derived length equal to 2. Furthermore, the bound is sharp.*

Proof. We argue by induction on the order of G. Note that G cannot have odd order according to Corollary 4.12. Furthermore, if H is a subgroup of G of odd order, then H must be abelian; otherwise, combining Corollary 4.12 with part (i) of Proposition 4.1, we would get

$$\frac{4}{81} < \mathrm{tp}(G) \leqslant \mathrm{tp}(H) \leqslant \frac{4}{81},$$

a contradiction. Also, since $\mathrm{tp}(G) > 4/81 > (1/2)^{14} = \mathrm{tp}(A_5)$, G is necessarily soluble by Proposition 4.3 in combination with Proposition 4.1 (iii).

Next, we argue that G has a unique minimal normal subgroup. Assume otherwise and let N_1, N_2 be distinct minimal normal subgroups of G. Since G/N_i is either abelian, or of derived length 2 by the induction hypothesis plus Proposition 4.1 (ii), we see that $(G/N_1)'$ is abelian, as is $(G/N_2)'$. We have

$$(G/N_i)' = G'N_i/N_i \cong G'/G' \cap N_i \quad (i = 1, 2)$$

and, as $G' \cap N_i \trianglelefteq G$, either $G' \cap N_i = 1$, or $N_i \leqslant G'$, due to minimality of N_i. If $G' \cap N_1 = 1$, then G' is abelian, and the induction is complete. Arguing similarly for N_2, we deduce that $N_1, N_2 \leqslant G'$. Therefore, both G'/N_1 and G'/N_2

are abelian groups. Now notice that $N_1 \cap N_2 = 1$, again due to minimality of the N_i. Moreover, the group $G'/(N_1 \cap N_2)$, which is isomorphic to a subgroup of $G'/N_1 \times G'/N_2$, is abelian as well. It follows that G' is abelian; thus, G has derived length 2, as required.

We may therefore assume that G has a unique minimal normal subgroup. It follows that, for every subgroup H of order 2 in G, either H is normal and thus central in G (and moreover, there can exist only one such subgroup by the previous observation) or, by part (i) of Proposition 4.1 plus Corollary 3.7, we have

$$2^{\frac{n-m}{2}} < \frac{81}{4},$$

where $n = |G|/2$ and $m = |N_G(H)|/2$. This implies $n - m \leqslant 8$ and thus

$$|G| - |N_G(H)| \leqslant 16.$$

Since $|N_G(H)| \leqslant |G|/2$, the previous inequality implies that $|G| \leqslant 32$. A GAP check confirms that every group of order at most 32 with $\mathrm{tp}(G) > 4/81$ has derived length at most 2, and thus the proof is complete in that case.

It therefore suffices to treat the case where a Sylow 2-subgroup of G has only one minimal subgroup which is moreover the unique minimal normal subgroup of G. If G is a 2-group, then it is either cyclic or a generalised quaternion group (these being the only 2-groups with a unique minimal subgroup); the first case is ruled out by our hypothesis that G is non-abelian. In the second case, the derived length equals 2. We may thus assume that G is not a 2-group.

Let P be a non-trivial Sylow p-subgroup of G for some odd prime p and let $T \leqslant P$ be of order p. As the only minimal normal subgroup of G is a 2-group, the cyclic group T is not normal in G; thus, $N_G(T) < G$ and, by Proposition 4.1 (i) and Corollary 3.7,

$$\frac{4}{81} < \mathrm{tp}(G) \leqslant \mathrm{tp}(T) \leqslant P_G(T) = \left(\frac{p!}{p^p}\right)^{\frac{n-m}{p}},$$

where $n = |G|/p$ and $m = |N_G(T)|/p < n$. Since

$$\frac{n-m}{p} \geqslant 1 \quad \text{and} \quad \frac{p!}{p^p} \leqslant \frac{24}{625} < \frac{4}{81}$$

for $p \geqslant 5$, we must have $p = 3$ and $n - m = 3$. Moreover, since $m \mid n$, we conclude that $m \mid 3$. Thus, either $m = 1$ and $|G| = 12$, or $m = 3$ and $|G| = 18$. As we have already checked, no group of order 12 or 18 has derived length greater than 2 and transversal probability greater than $4/81$, and our proof is complete.

Finally, note that the transversal probability of $\mathrm{SL}_2(3)$ equals $4/81$ by part (ix) of Lemma 4.2, while its derived length is 3, so that our bound is indeed sharp, as claimed. □

4.3 Group extensions and the function tp(−)

The aim in this section is to see how the function tp(−) behaves with respect to group extensions.

Let G and K be groups, and let (V, W) be a factor system for G by K. In detail, this means that $V\colon K \to \mathrm{Aut}(G)$ and $W\colon K \times K \to G$ are maps such that

$$V(k_2) \circ V(k_1) = i_{W(k_1,k_2)} \circ V(k_1k_2) \quad (k_1, k_2 \in K), \tag{4.6}$$

$$W(k_1, k_2k_3)W(k_2, k_3) = W(k_1k_2, k_3)V(k_3)(W(k_1, k_2)) \quad (k_1, k_2, k_3 \in K), \tag{4.7}$$

where $i_g(x) = g^{-1}xg$ for $x, g \in G$, so that i_g is the inner automorphism of G associated with the element $g \in G$. Let $\hat{G}$ be the extension of G by K associated with (V, W), that is, $\hat{G} = K \times G$ as a set, with group law given by

$$(k_1, g_1) \cdot (k_2, g_2) = (k_1k_2, W(k_1, k_2)V(k_2)(g_1)g_2) \quad (k_1, k_2 \in K,\ g_1, g_2 \in G), \tag{4.8}$$

and we have a short exact sequence

$$1 \to G \xrightarrow{\iota} \hat{G} \xrightarrow{\pi} K \to 1,$$

where

$$\iota(g) = (1, W(1, 1)^{-1}g) \quad \text{and} \quad \pi(k, g) = k,$$

and the associated map $\varphi_E\colon K \to \hat{G}$ is given by $\varphi_E(k) = (k, 1)$; cf. [22, Section 9.4, in particular Statement 9.4.5]. We shall need the following properties of a factor system.

Lemma 4.15. *Let (V, W) be a factor system for G by K. Then*

(i) $V(1) = i_{W(1,1)}$,

(ii) $W(k, 1) = W(1, 1)$ *($k \in K$),*

(iii) $W(1, k) = V(k)(W(1, 1))$ *($k \in K$).*

Proof. (i) Setting $k_1 = k_2 = 1$ in (4.6) gives $V(1) \circ V(1) = i_{W(1,1)} \circ V(1)$. Thus,

$$V(1) = V(1) \circ V(1) \circ V(1)^{-1} = i_{W(1,1)} \circ V(1) \circ V(1)^{-1} = i_{W(1,1)},$$

as desired.

(ii) Setting $k_1 = k$ and $k_2 = k_3 = 1$ in (4.7), we get

$$W(k, 1)W(1, 1) = W(k, 1)V(1)(W(k, 1)).$$

Applying part (i), this equation may be rewritten as

$$W(k,1)W(1,1) = W(k,1)W(1,1)^{-1}W(k,1)W(1,1).$$

Multiplying the last equation from the left by $W(k,1)^{-1}$ and from the right by $W(1,1)^{-1}$, the result follows.

(iii) Setting $k_1 = k_2 = 1$ and $k_3 = k$ in (4.7) yields

$$W(1,k)^2 = W(1,k)V(k)(W(1,1)),$$

whence (iii). □

Theorem 4.16. *Let $\hat{G}$ be an extension of the group G by the group K with associated factor system (V,W). Then*

$$\mathrm{tp}(\hat{G}) \leqslant \min_{H \leqslant G} \prod_{k \in K} P_G(H, V(k)(H)). \tag{4.9}$$

Proof. Let $H \leqslant G$ be a subgroup (with G considered as a subgroup of $\hat{G}$ via the embedding ι). Then, for $h \in H$ and $(k,g) \in \hat{G}$, we have

$$\begin{aligned}(k,g)\iota(h) &= (k,g)(1, W(1,1)^{-1}h)\\ &= (k, W(k,1)V(1)(g)W(1,1)^{-1}h)\\ &= (k, W(k,1)W(1,1)^{-1}gW(1,1)W(1,1)^{-1}h)\\ &= (k, W(k,1)W(1,1)^{-1}gh) = (k,gh),\end{aligned}$$

where we have applied the definition of ι in the first step, (4.8) in the second step, Lemma 4.15 (i) in the third step, and Lemma 4.15 (ii) in the last step. Hence,

$$(k,g)\cdot H = (k,gH) \quad ((k,g) \in \hat{G},\ H \leqslant G). \tag{4.10}$$

Similarly, making use of the definition of ι, the group law (4.8), and part (iii) of Lemma 4.15, we find that, for $h \in H$ and $(k,g) \in \hat{G}$,

$$\begin{aligned}\iota(h)(k,g) &= (1, W(1,1)^{-1}h)(k,g)\\ &= (k, W(1,k)V(k)(W(1,1)^{-1}h)g)\\ &= (k, W(1,k)V(k)(W(1,1))^{-1}V(k)(h)g)\\ &= (k, V(k)(h)g),\end{aligned}$$

implying

$$H\cdot(k,g) = (k, V(k)(H)g) \quad ((k,g) \in \hat{G},\ H \leqslant G). \tag{4.11}$$

Now let

$$\tilde{T} = \{(k_i, g_j)\}_{(i,j)} \in \mathrm{DT}_{\hat{G}}(H)$$

be any two-sided transversal for H in $\hat{G}$. From either (4.10) or (4.11), when applied to $H \leqslant G$ and $(k_i, g_j) \in \tilde{T}$, it is clear that *all* elements of K must occur as first component of an element in $\tilde{T}$, since otherwise

$$(k, 1) \notin \bigcup_{(i,j)} (k_i, g_j)H = \hat{G}$$

for some $k \in K$, a contradiction. Given $k \in K$, consider the set

$$T(k) := \{g \in G : (k, g) \in \tilde{T}\} \subseteq G.$$

We claim that $T(k) \in \mathrm{DT}_G(H, V(k)(H))$.

First, let $k \in K$ be given, and let $g \in G$ be arbitrary. Then

$$(k, g) \in (k_i, g_j)H = (k_i, g_j H)$$

for some (i, j). This implies $k_i = k$ and $g \in g_j H$. Thus,

$$g_j \in T(k) \quad \text{and} \quad g \in \bigcup_{g_\ell \in T(k)} g_\ell H$$

so that

$$G = \bigcup_{g_\ell \in T(k)} g_\ell H.$$

Similarly, given $g \in G$, we have

$$(k, g) \in H(k_i, g_j) = (k_i, V(k_i)(H)g_j)$$

for some (i, j); thus, $k_i = k$, $g \in V(k)(H)g_j$, and $g_j \in T(k)$ so that

$$G = \bigcup_{g_\ell \in T(k)} V(k)(H)g_\ell.$$

Next, suppose that $g_j H \cap g_\ell H \neq \emptyset$, where $g_j, g_\ell \in T(k)$. Then

$$(k, g_j)H \cap (k, g_\ell)H = (k, g_j H) \cap (k, g_\ell H) = (k, g_j H \cap g_\ell H) \neq \emptyset,$$

implying $(k, g_j) = (k, g_\ell)$ by our hypothesis on $\tilde{D}$; thus, $g_j = g_\ell$. Hence, $T(k)$ is a left transversal for H in G. Similarly, suppose that

$$V(k)(H)g_j \cap V(k)(H)g_\ell \neq \emptyset$$

for some $g_j, g_\ell \in T(k)$. Then we have

$$\begin{aligned} H(k, g_j) \cap H(k, g_\ell) &= (k, V(k)(H)g_j) \cap (k, V(k)(H)g_\ell) \\ &= (k, V(k)(H)g_j \cap V(k)(H)g_\ell) \neq \emptyset \end{aligned}$$

so that, again, $g_j = g_\ell$. Consequently, $T(k)$ is also a right transversal for $V(k)(H)$ in G; therefore, $T(k) \in \mathrm{DT}_G(H, V(k)(H))$, as claimed.

Mapping $k \in K$ to $T(k)$ for given $\tilde{T}$ thus gives a choice function

$$f(\tilde{T})\colon K \to \bigsqcup_{k \in K} \mathrm{DT}_G(H, V(k)(H))$$

and, subsequently, sending $\tilde{T} \in \mathrm{DT}_{\hat{G}}(H)$ to $f(\tilde{T})$, defines a map

$$\tilde{\Phi}\colon \mathrm{DT}_{\hat{G}}(H) \to \mathrm{CF}(K, H, G),$$

where, unsurprisingly, $\mathrm{CF}(K, H, G)$ denotes the set of all choice functions

$$f\colon K \to \bigsqcup_{k \in K} \mathrm{DT}_G(H, V(k)(H)),$$

that is, functions f as above such that $f(k) \in \mathrm{DT}_G(H, V(k)(H))$ for each $k \in K$. For later use, we observe that, obviously,

$$|\mathrm{CF}(K, H, G)| = \prod_{k \in K} |\mathrm{DT}_G(H, V(k)(H))|.$$

Next, we note that if $\tilde{T}_1, \tilde{T}_2 \in \mathrm{DT}_{\hat{G}}(H)$ are such that $\tilde{\Phi}(\tilde{T}_1) = f = \tilde{\Phi}(\tilde{T}_2)$, then by definition of $\tilde{\Phi}$,

$$\tilde{T}_1 = \bigcup_{k \in K} (k, f(k)) = \tilde{T}_2$$

so that $\tilde{\Phi}$ is injective. We want to show that $\tilde{\Phi}$ is surjective as well. Let

$$f \in \mathrm{CF}(K, H, G)$$

be given, and set

$$\tilde{T}_f := \bigcup_{k \in K} (k, f(k)) \subseteq \hat{G}.$$

We claim that $\tilde{T}_f \in \mathrm{DT}_{\hat{G}}(H)$. By (4.10), we have

$$\begin{aligned} \bigcup_{\tilde{t} \in \tilde{T}_f} \tilde{t} H &= \bigcup_{k \in K} \bigcup_{g \in f(k)} (k, g)H = \bigcup_{k \in K} \bigcup_{g \in f(k)} (k, gH) \\ &= \bigcup_{k \in K} \Big(k, \bigcup_{g \in f(k)} gH\Big) = \bigcup_{k \in K} (k, G) = \hat{G}. \end{aligned}$$

Similarly, by (4.11), we have

$$\bigcup_{\tilde{t}\in\tilde{T}_f} H\tilde{t} = \bigcup_{k\in K}\bigcup_{g\in f(k)} H(k,g) = \bigcup_{k\in K}\bigcup_{g\in f(k)} (k, V(k)(H)g)$$
$$= \bigcup_{k\in K}\Big(k, \bigcup_{g\in f(k)} V(k)(H)g\Big) = \bigcup_{k\in K}(k,G) = \hat{G}.$$

Moreover, for $\tilde{t}_1 = (k_1, g_1), \tilde{t}_2 = (k_2, g_2) \in \tilde{T}_f$, where $k_1, k_2 \in K$, $g_1 \in f(k_1)$, and $g_2 \in f(k_2)$, the hypothesis

$$\tilde{t}_1 H \cap \tilde{t}_2 H = (k_1, g_1)H \cap (k_2, g_2)H = (k_1, g_1 H) \cap (k_2, g_2 H) \neq \emptyset$$

first implies $k_1 = k_2 =: k$, and thus $g_1, g_2 \in f(k)$, as well as $g_1 H \cap g_2 H \neq \emptyset$, which forces $g_1 = g_2$, since $f(k)$ forms a left transversal for H in G. Hence, $\tilde{t}_1 = \tilde{t}_2$. Also, with $\tilde{t}_1, \tilde{t}_2, k_1, k_2, g_1, g_2$ as above, the assumption that

$$H\tilde{t}_1 \cap H\tilde{t}_2 = H(k_1, g_1) \cap H(k_2, g_2)$$
$$= (k_1, V(k_1)(H)g_1) \cap (k_2, V(k_2)(H)g_2) \neq \emptyset$$

implies $k_1 = k_2 =: k$; thus, $g_1, g_2 \in f(k)$, and $V(k)(H)g_1 \cap V(k)(H)g_2 \neq \emptyset$, forcing $g_1 = g_2$, as $f(k)$ also forms a right transversal for $V(k)(H)$ in G. Hence, again, $\tilde{t}_1 = \tilde{t}_2$, and it follows that $\tilde{T}_f \in \mathrm{DT}_{\hat{G}}(H)$. Since $\tilde{\Phi}(\tilde{T}_f) = f$ by construction, we conclude that $\tilde{\Phi}$ is surjective, thus a bijection. Therefore,

$$|\mathrm{DT}_{\hat{G}}(H)| = |CF(K, H, G)|,$$

and we conclude that

$$\mathrm{tp}(\hat{G}) \leqslant \min_{H\leqslant G} P_{\hat{G}}(H) = \min_{H\leqslant G} \frac{|\mathrm{DT}_{\hat{G}}(H)|}{|H|^{(\hat{G}:H)}} = \min_{H\leqslant G} \frac{|\mathrm{DT}_{\hat{G}}(H)|}{|H|^{(G:H)|K|}}$$
$$= \min_{H\leqslant G} \prod_{k\in K} \frac{|\mathrm{DT}_G(H, V(k)(H))|}{|H|^{(G:H)}}$$
$$= \min_{H\leqslant G} \prod_{k\in K} P_G(H, V(k)(H)),$$

whence (4.9). □

Corollary 4.17. *Let $\hat{G} = G \rtimes K$, viewed as an internal semidirect product. Then we have*

$$\mathrm{tp}(\hat{G}) \leqslant \min_{H\leqslant G} \prod_{k\in K} P_G(H, H^k).$$

Proof. Suppose that the factor system (V, W) in Theorem 4.16 splits so that (V, W) is equivalent to some factor system (V^*, W^*) such that $V^*\colon K \to \mathrm{Aut}(G)$ is a homomorphism, $W^*(k_1, k_2) = 1$ for all $k_1, k_2 \in K$, and φ_E is a section to the projection π. Identifying $g \in G$ with $\iota(g) = (1, g)$ and $k \in K$ with $\varphi_E(k) = (k, 1)$, we have

$$V(k)(g) = g^k \quad (k \in K,\ g \in G)$$

so that $\mathrm{DT}_G(H, V(k)(H)) = \mathrm{DT}_G(H, H^k)$, and thus

$$P_G(H, V(k)(H)) = P_G(H, H^k).$$

Our claim follows now from Theorem 4.16. □

Corollary 4.18. *Let $\hat{G} = G_1 \times \cdots \times G_r$, where the G_ρ are finite groups, and set $m := |\hat{G}|$. Then we have*

$$\mathrm{tp}(\hat{G}) \leqslant \min_{1 \leqslant \rho \leqslant r} \mathrm{tp}(G_\rho)^{m/|G_\rho|}. \tag{4.12}$$

Proof. An immediate induction on r, starting with the trivial case where $r = 1$, reduces us to the case where $r = 2$, that is, to

$$\mathrm{tp}(G \times K) \leqslant \min\{\mathrm{tp}(G)^{|K|}, \mathrm{tp}(K)^{|G|}\}. \tag{4.13}$$

Moreover, as tp is an isomorphism invariant by Corollary 3.16, it suffices to show for (4.12) that $\mathrm{tp}(G \times K) \leqslant \mathrm{tp}(G)^{|K|}$. However, assuming that $[G, K] = 1$, Corollary 4.17 gives

$$\mathrm{tp}(G \times K) \leqslant \min_{H \leqslant G} \prod_{k \in K} P_G(H) = \prod_{k \in K} \min_{H \leqslant G} P_G(H) = \mathrm{tp}(G)^{|K|},$$

whence the result. □

It is easy to construct examples where inequality (4.12) is sharp. For instance, let $G = S_3 \times C_p$, where $p \geqslant 5$ is a prime number. Then the minimum on the right-hand side of (4.13) equals $\mathrm{tp}(S_3)^p = 2^{-p}$. Since $p \nmid |S_3|$, all subgroups of G are of the form $H = U \times C$, where $U \leqslant S_3$ and $C \leqslant C_p$. Discarding normal subgroups and making use of Corollary 3.16, we see that, as regards $\mathrm{tp}(G)$, we only need to check the subgroups $H_1 = \langle (1, 2) \rangle \times 1$ and $H_2 = \langle (1, 2) \rangle \times C_p$, whose transversal probabilities are given by $P_G(H_1) = 2^{-p}$ and $P_G(H_2) = 1/2$, respectively. Hence,

$$\mathrm{tp}(G) = \min\{1, 1/2, 2^{-p}\} = 2^{-p},$$

as desired.

5 Some problems and questions

In this final section, we outline several open problems, conjectures, and questions, hoping thereby to stimulate further research on this topic.

Problem 5.1. Assume a block diagonal doubly stochastic matrix

$$T = \operatorname{diag}(T_1, \ldots, T_s)$$

is given, where, for $i = 1, \ldots, s$, the matrix T_i has order t_i, and its entries all equal $1/t_i$. Do there exist a finite group G and subgroups H, K of G (of the same order) such that $\operatorname{per}(T) = P_G(H, K)$? If, in addition, we know that the number of t's equal to 1 is non-zero, do there exist a finite group G and a subgroup H such that $\operatorname{per}(T) = P_G(H)$?

Problem 5.2. Suppose $H \leqslant G$ and $N \trianglelefteq G$, where G is finite. Is it then always true that $P_G(H) \leqslant P_G(HN)$ (GAP computations confirm this up to $|G| = 200$)? Such a result would have an interesting consequence: combining it with Theorem 3.15, it would follow that, for any group homomorphism $f: G \to K$, and with $N := \ker(f)$,

$$P_G(H) \leqslant P_G(HN) = P_{f(G)}(f(HN)) = P_{f(G)}(f(H)) \tag{5.1}$$

so that we would have a general inequality relating to the function $P(-)$ associated with each homomorphism f. Moreover, it should be possible to characterise equality in (5.1).

Problem 5.3. Characterise equality in (4.13). In particular, equality should hold if $(|G|, |K|) = 1$ (this seems plausible, and is supported by massive computational evidence).

Problem 5.4. What is the appropriate setting for the quantity $P_G(H)$ to make sense when G is an infinite group and H is a finite index subgroup of G?

Recall that $\operatorname{cp}(G)$ is the commuting probability of G, i.e., the probability that two randomly chosen elements of G commute. Regarding the relation between $\operatorname{tp}(G)$ and $\operatorname{cp}(G)$, we propose the following conjecture.

Problem 5.5. Let G be a finite group. Then we have $\operatorname{tp}(G) \leqslant \operatorname{cp}(G)$ except if G is Dedekind non-abelian, in which case $\operatorname{cp}(G)/\operatorname{tp}(G) = 5/8$, or $G = Q_{16}$ in which case $\operatorname{cp}(G)/\operatorname{tp}(G) = 7/8$. Moreover, if $\operatorname{cp}(G) = \operatorname{tp}(G)$, then either G is abelian or

$$G \cong \langle a, b : a^3 = b^{2^n} = 1, a^b = a^{-1} \rangle$$

for some positive integer n, and thus $\operatorname{tp}(G) = \operatorname{cp}(G) = 1/2$.

Problem 5.6. Does Theorem 4.6 generalise? In particular, is the following true? Assume that

$$\prod_{i=1}^{n} f(t_i) = \prod_{j=1}^{k} f(s_j),$$

where t_i, s_j are distinct integers greater than 1, for all appropriate i, j and

$$f(x) = \Gamma(x+1)/x^x.$$

Can we then conclude that $n = k$ and $t_i = s_i$ for all i after rearranging appropriately?

Problem 5.7. Let G be a finite group and suppose that $K \leqslant H \leqslant G$. In Proposition 3.14, we saw that $P_G(K) \leqslant P_H(K)$. Are $P_G(K)$ and $P_G(H)$ connected in some way? If so, how?

Finally, as regards the quantity $\mathrm{tp}(G)$, we ask the following which, in our opinion, is the most important relevant question.

Problem 5.8. Let G be a finite group. Does there exist a prime divisor p of $|G|$ and a cyclic p-subgroup H such that $\mathrm{tp}(G) = P_G(H)$ and moreover H has the property $(H : N) \leqslant p$, where $N = \mathrm{core}_G(H)$?

We think the answer is "yes".

A Majorisation

We begin by recalling basic definitions and concepts from the theory of majorisation and refer the reader to the canonical work on this topic [19] for further information. Towards the end of this section, we will prove a result that we have appealed to in the proof of Theorem 4.6.

Fix a positive integer s and let $\mathbb{R}_+ := [0, +\infty)$. For any $x = (x_1, \ldots, x_s) \in \mathbb{R}^s$, let $x_{[1]} \geqslant \cdots \geqslant x_{[s]}$ denote the components of x in decreasing order, and let

$$x_\downarrow = (x_{[1]}, \ldots, x_{[s]})$$

denote the decreasing rearrangement of x.

Definition A.1. For $x, y \in \mathbb{R}^s$, we write $x \prec y$ and say that x is majorised by y (or that y majorises x) if

$$\sum_{i=1}^{k} x_{[i]} \leqslant \sum_{i=1}^{k} y_{[i]}, \quad k = 1, \ldots, s-1, \quad \text{and} \quad \sum_{i=1}^{s} x_{[i]} = \sum_{i=1}^{s} y_{[i]}.$$

Inequality in the final equality in the definition above leads to the concept of weak majorisation.

Definition A.2. For $x, y \in \mathbb{R}^n$, we write $x \prec_{\mathrm{w}} y$ and say that x is weakly majorised by y (or that y weakly majorises x) if, for all $k \in [s]$,

$$\sum_{i=1}^{k} x_{[i]} \leqslant \sum_{i=1}^{k} y_{[i]}.$$

We present now the fundamental concept of Schur-convexity/concavity.

Definition A.3. A real-valued function ϕ defined on a set $\mathcal{A} \subset \mathbb{R}^n$ is said to be Schur-convex on $\mathcal{A}$ if $x \prec y$ on $\mathcal{A}$ implies that $\phi(x) \leqslant \phi(y)$. If, in addition, $\phi(x) < \phi(y)$ whenever $x \prec y$ but x is not a permutation of y, then ϕ is said to be strictly Schur-convex on $\mathcal{A}$. Similarly, ϕ is said to be Schur-concave on $\mathcal{A}$ if $x \prec y$ on $\mathcal{A}$ implies that $\phi(x) \geqslant \phi(y)$ and ϕ is strictly Schur-concave on $\mathcal{A}$ if strict inequality $\phi(x) > \phi(y)$ holds when x is not a permutation of y.

Schur's fundamental result asserts that if ϕ is a convex real function, then the function $\sum_{i=1}^{s} \phi(x_i)$ is Schur-convex. Since we are interested in products, the following result helps make the transition from sums to products.

Theorem A.4 ([19, Proposition E1, pp. 105–106]). *Let f be a continuous non-negative function defined on an interval $I \subset \mathbb{R}$. Then*

$$h(x) = \prod_{i=1}^{s} f(x_i), \quad x \in I^s,$$

is Schur-convex/concave on I^s if and only if $\log f$ *is convex/concave on I. Moreover, h is strictly Schur-convex/concave on I^s if and only if* $\log f$ *is strictly convex/concave on I.*

Of course, it follows from the above result that $\phi(x) = \prod_{i=1}^{n} g(x_i)$ is (strictly) Schur-concave if and only if $\log g$ is (strictly) concave.

Finally, we mention the following crucial result.

Theorem A.5 ([19, Statement A.9.a, p. 177]). *Let $x, y \in \mathbb{R}^s$. If $x \prec_{\mathrm{w}} y$, then there exists a vector v such that $x \prec v$ and $v \leqslant y$, that is, we have $v_i \leqslant y_i$ for all $i \in [s]$.*

For $x \geqslant 0$, recall that

$$f(x) := \frac{\Gamma(x+1)}{x^x},$$

where $\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}\,dt$ is the Γ-function. Note that $\log f$ is strictly concave on $I = [0, +\infty)$, owing to the fact that $(\log f)'' < 0$, which we have established in Section 3.4.

It is a consequence of Theorem A.4 and the comment immediately following it that, since $\log f(x)$ is strictly concave on $I = [0, +\infty)$, the function $h\colon I^s \to \mathbb{R}$ with

$$h(x_1, \dots, x_s) = \prod_{\sigma=1}^{s} f(x_\sigma) \tag{A.1}$$

is strictly Schur-concave on I^s.

Proposition A.6. *Let $x, y \in I^s$ and suppose that $x = x_\downarrow$, $y = y_\downarrow$. Assume that $x \prec_{\mathrm{w}} y$, that is, $x_1 + \cdots + x_i \leqslant y_1 + \cdots + y_i$ for all $i \in [s]$. Suppose that the coordinates of x are pairwise distinct. Moreover, assume that there exists a $k \in [s]$ such that $x_1 + \cdots + x_j < y_1 + \cdots + y_j$ for all $j \in \{k, \dots, s\}$, so that strict inequality holds in the weak majorisation order between x and y from a certain point on. Then $h(x) > h(y)$, where h is as in (A.1).*

Proof. By Theorem A.5, there exists a vector v such that $x \prec v$ and $v \leqslant y$. Since $x \prec v$ and h is strictly Schur-concave, we have that $h(x) > h(v)$. We justify this claim. We certainly have $h(x) \geqslant h(v)$. If we had $h(x) = h(v)$, then (by definition) v would have to be a permutation of x. But x has distinct coordinates, so we would have $x = v$ and thus $x_i \leqslant y_i$ for all i. Now f is strictly decreasing for $x \geqslant 1$; thus, $f(x_i) \geqslant f(y_i)$, and moreover, at least one strict inequality holds by the initial assumption on k. Therefore,

$$h(x) = h(x_1, \dots, x_s) = \prod_{\sigma=1}^{s} f(x_\sigma) > \prod_{\sigma=1}^{s} f(y_\sigma) = h(y),$$

as wanted.

We may therefore assume that $h(x) > h(v)$. An argument similar to above shows that $v \leqslant y$ plus monotonicity of f implies that $h(v) \geqslant h(y)$. In conclusion, we have $h(x) > h(v) \geqslant h(y)$, proving our assertion. □

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